

Seaglider SG194

Mass and Inertia Matrix Estimation

Authors: Geenadie Rathnayake and Holland Kantner

NOTE

This analysis was performed to estimate the added mass and inertia matrices as well as stationary inertia matrix. This analysis is specific to the SG 194 on the day of October 23rd, 2013 ballasting date since Trim and Balance sheet from this day was used for this analysis. However, these same steps can be performed for any glider with the same shape. Note that properties of all components such as mass, longitudinal center of gravity, vertical center of gravity, aspect ratio of the glider may vary from glider to glider. If using a different glider from SG 194, use the steps outlined in this document to estimate the mass and inertia matrices.

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Assumptions

#	Assumption	Impact
1	Prolate spheroid hull	Wings, antenna, and sensor fairings are neglected. Assuming three planes of symmetry. Standard practice for slender underwater vehicle dynamics. Affects $\mathbf{M}_f$ and $\mathbf{J}_f$.
2	Potential flow	Brennen Table B5 (3) is valid for potential flow, and viscous flow at that first moment when the body and fluid are accelerating from rest. Other viscous effects are not captured. Affects $\mathbf{M}_f$ and $\mathbf{J}_f$
3	Lateral CG latCG = y_{CG} = 0 for all components except wings	All components are located along the centerline from nose to tail (x) and only shifted up or down (z). However, this is NOT TRUE for wings since wings are located away from vehicle CG in lateral direction (y). This is important when calculating stationary inertia matrices for damage modes. Port-starboard symmetry assumed for all components. Small real asymmetries are neglected. Affects $\mathbf{J}_s$.
4	d_z uses vehicle VCG	VCG = -0.835 cm is a vehicle-level value. Per-component vertical offsets are not listed in the spreadsheet.
5	Vehicle CG shift due to faults neglected	An analysis of shift in center of gravity (CG) of the glider due to pre-defined faults was performed and determined that CG shifts are negligible.

Coordinate Frame

The origin is placed at the vehicle center of gravity (CG). All distances are measured from this point.

Axis	Definition
x	Along the hull axis, positive toward the nose
y	Lateral, positive to starboard
z	Vertical, positive upward

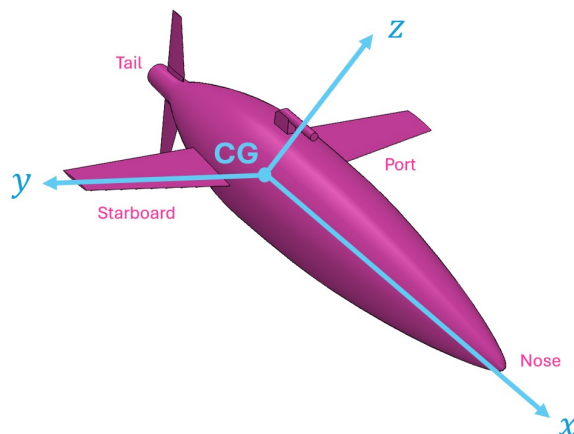


Figure 1: Coordinate system at CG

Step 1 — Vehicle Parameters from Spreadsheet

The following values are read directly from the SG194 trim and balance spreadsheet summary (Page 1):

Parameter	Value	Source in spreadsheet
Total mass of glider m_{total}	58852 g = 58.852 kg	Total Scale Weight row
Longitudinal CG x_{CG} (LCG)	95.752 cm = 0.9575 m from nose	“Longitudinal CG” row
Vertical CG z_{CG} (VCG)	−0.835 cm = −0.00835 m from neutral axis	“Vertical CG (VCG)” row
Lateral CG y_{CG}	0.00 m	Assumption, not mentioned in the data sheet
Water density ρ	1025 kg/m ³	1.02730 g/cc at 3.5°C, 1000 dbar

Note on LCG: The LCG (Longitudinal Center of Gravity) values in the spreadsheet give the x -position of each component measured along the hull axis from the nose. The vertical position (z -direction) is captured by the VCG (Vertical Center of Gravity). The LatCG (Lateral Center of Gravity) is not given and assumed in the spreadsheet.

Step 2 — Movable Battery Mass

Battery mass

The mass shifter assembly (part 49889) contains three components that translate forward and aft along the hull to control pitch. The LV battery is bolted to the static electronics rail and does not move.

Component	Part #	Mass (g)	Mass (kg)	Status
HV Battery	52238	9785.0	9.7850	Movable
Brass weight		1372.6	1.3726	Movable
Battery enclosure	56759	478.0	0.4780	Movable
Total movable m_b		11635.6	11.6356	
LV Battery pack (static)	52237	2664.4	2.6644	Not included

$$m_b = 9.7850 + 1.3726 + 0.4780 = \boxed{11.6356 \text{ kg}}$$

Battery LCG

The longitudinal position of the movable battery CG is the mass-weighted average of its three components, using LCG values read from the spreadsheet:

$$\text{LCG, bat} = \frac{\sum m_\alpha x_\alpha}{\sum m_\alpha} = \frac{(9.7850 \times 90.8) + (1.3726 \times 90.0) + (0.4780 \times 90.0)}{11.6356} = 90.673 \text{ cm}$$

$$\text{LCG, bat} = 0.9067 \text{ m from nose}$$

Battery VCG

The vertical position of the movable battery CG is the mass-weighted average of its three components, using VCG values read from the spreadsheet:

$$\text{VCG, bat} = \frac{\sum m_\alpha z_\alpha}{\sum m_\alpha} = \frac{(9.7850 \times 0) + (1.3726 \times -6.80) + (0.4780 \times -6.80)}{11.6356} = -1.082 \text{ cm}$$

$$\text{VCG, bat} = -0.01082 \text{ m from neutral axis}$$

Battery CG offset from vehicle CG

Direction	Formula	Calculation	Result	Notes
$x_{CG,bat}$	$\text{LCG, bat} - x_{CG}$	$0.9067 - 0.9575$	-0.0508 m	Forward of vehicle CG
$y_{CG,bat}$	Bilateral symmetry		0 m	
$z_{CG,bat}$	$\text{VCG, bat} - z_{CG}$	$-0.01082 - (-0.00835) \text{ m}$	-0.00247 m	Below vehicle CG

Step 3 — Hull Geometry

The Seaglider hull is modelled as a **prolate spheroid** (an ellipsoid elongated along one axis). These values were obtained from the CAD model of the Seaglider.

Dimension	full length	Semi-axis
Major axis	1.80 m	$a = 1.80/2 = 0.90$ m
Minor axis	0.28428 m	$b = 0.28428/2 = 0.14214$ m
Aspect ratio a/b		$0.90/0.14214 = 6.332$

Step 4 — Added Mass Matrix $\mathbf{M}_f$

When the glider accelerates, it must also accelerate the surrounding fluid, which acts like extra inertia. This is captured by the added mass matrix:

$$\mathbf{F} = -\mathbf{M}_f \mathbf{a}$$

For a prolate spheroid, $\mathbf{M}_f$ is diagonal because three planes of symmetry prevent cross-coupling between axes.

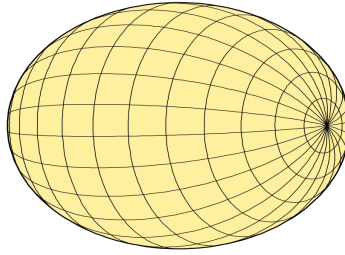


Figure 2: Prolate spheroid

Step 4.1 — Interpolate Brennen Table B5

The Brennen table gives non-dimensional coefficients M_{11} , M_{22} , M_{55} at specific a/b values. Aspect ratio $a/b = 6.33$ falls between the table entries at $a/b = 6.01$ (point A) and $a/b = 8.01$ (point B).

The linear interpolation formula used is: $M = M_A + \text{slope} \left(\left(\frac{a}{b} \right) - \left(\frac{a}{b} \right)_A \right)$ where

$$\text{slope} = \frac{M_A - M_B}{\left(\frac{a}{b} \right)_A - \left(\frac{a}{b} \right)_B} \quad (1)$$

k	Coefficient	(A) $a/b =$ 6.01	(B) $a/b =$ 8.01	slope	Interpolated at $a/b = 6.33$
k_{11}	$M_{11}/\frac{4}{3}\rho\pi ab^2$	0.045	0.029	-0.008	$0.045 + -0.008 \times (6.33 - 6.01) = 0.0424$
k_{22}	$M_{22}/\frac{4}{3}\rho\pi ab^2$	0.918	0.945	0.0135	$0.918 + 0.1609 \times (6.33 - 6.01) = 0.9223$
k_{55}	$M_{55}/\frac{4}{15}\rho\pi ab^2(a^2 + b^2)$	0.764	0.840	0.038	$0.764 + 0.1609 \times (6.33 - 6.01) = 0.7762$

Two reference quantities that the coefficients scale against:

$$G_{\text{translation}} = \frac{4}{3} \pi \rho a b^2 = \frac{4}{3} \times \pi \times 1025 \frac{\text{kg}}{\text{m}^3} \times 0.90\text{m} \times (0.14214)^2 \text{m}^2 = 78.071 \text{ kg} \quad (2)$$

$$G_{\text{rotational}} = \frac{4}{15} \pi \rho a b^2 (a^2 + b^2) = \frac{4}{15} \times \pi \times 1025 \frac{\text{kg}}{\text{m}^3} \times 0.90\text{m} \times (0.14214)^2 \text{m}^2 \times (0.9^2 + 0.14214^2) \text{m} \quad (3)$$

$$G_{\text{rotational}} = 12.963 \text{ kg m}^2 \quad (4)$$

Step 4.2 — Added mass matrix assembly

M_{nn}	Formula	Calculation	Result
M_{11}	$k_{11} \times G_{\text{translation}}$	0.0424×78.071	3.310 kg
M_{22}	$k_{22} \times G_{\text{translation}}$	0.9223×78.071	72.005 kg
M_{33}	$M_{33} = M_{22}$		72.005 kg
Off-diagonals	0 (3 planes of symmetry)		0 kg m ²

$M_{11} \ll M_{22}$: moving nose-first the slender hull displaces very little fluid. Moving sideways it must push a large volume of water out of the way.

$$\mathbf{M}_f = \begin{bmatrix} M_{11} & 0 & 0 \\ 0 & M_{22} & 0 \\ 0 & 0 & M_{33} \end{bmatrix} = \begin{bmatrix} 3.310 & 0 & 0 \\ 0 & 72.005 & 0 \\ 0 & 0 & 72.005 \end{bmatrix} \text{ kg}$$

Step 5 — Added Inertia Matrix $\mathbf{J}_f$

The added inertia matrix is the rotational equivalent of added mass:

$$\mathbf{T} = -\mathbf{J}_f \boldsymbol{\alpha}$$

Entry	Formula	Result
J_{xx} (roll)	(body of revolution spinning about its own symmetry axis does not drag surrounding fluid)	0 kg m ²
J_{yy} (pitch)	$k_{55} \times G_{\text{rotational}} = 0.7762 \times 12.963$	10.062 kg m ²
J_{zz} (yaw)	$J_{zz} = J_{yy}$	10.062 kg m ²
Off-diagonals		0 kg m ²

Step 5.1 — Added inertia matrix assembly

$$\mathbf{J}_f = \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_{A,yy} & 0 \\ 0 & 0 & I_{A,zz} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 10.062 & 0 \\ 0 & 0 & 10.062 \end{bmatrix} \text{ kg m}^2$$

Step 6 — Stationary Inertia Matrix $\mathbf{J}_s$ **Step 6.1 — Gather data needed spreadsheet**

For each component listed in the Trim and Balance sheet of SG 194 (1), any components with a recorded mass of 0 g and movable battery pack (HV battery, Brass weight, Battery enclosure) were omitted from the calculation since this is only for stationary components. For the remaining components, the mass was converted to kilograms, and the longitudinal center of gravity (LCG) and vertical center of gravity (VCG) were recorded in centimeters. The lateral center of gravity (LatCG) of wings was obtained from the CAD. For all other components, LatCG was assumed to be zero because it was not provided in the trim sheet. Using these values, the component offsets from the reference center of gravity were calculated as Δx , Δz , and Δy using the following formulas.

$$\begin{aligned}\Delta x &= \text{LCG} - x_{CG} \\ \Delta z &= \text{VCG} - z_{CG} \\ \Delta y &= \text{latCG} - y_{CG}\end{aligned}$$

where $x_{CG} = 95.752$ cm, $z_{CG} = -0.835$ cm, and $y_{CG} = 0.00$ cm.

The complete list of components used in this step is in Appendix A.

Step 6.2 — Methodology

From Cline (2) in the references, the diagonal moments of inertia of the inertia tensor in cartesian coordinates are:

$$J_{xx} = \sum_{\alpha}^N m_{\alpha} (\Delta y_{\alpha}^2 + \Delta z_{\alpha}^2) \quad (5)$$

$$J_{yy} = \sum_{\alpha}^N m_{\alpha} (\Delta x_{\alpha}^2 + \Delta z_{\alpha}^2) \quad (6)$$

$$J_{zz} = \sum_{\alpha}^N m_{\alpha} (\Delta x_{\alpha}^2 + \Delta y_{\alpha}^2) \quad (7)$$

while the off-diagonal products of inertia are

$$J_{yx} = I_{xy} = - \sum_{\alpha}^N m_{\alpha} (\Delta x_{\alpha} \Delta y_{\alpha}) \quad (8)$$

$$J_{zx} = I_{xz} = - \sum_{\alpha}^N m_{\alpha} (\Delta x_{\alpha} \Delta z_{\alpha}) \quad (9)$$

$$J_{zy} = I_{yz} = - \sum_{\alpha}^N m_{\alpha} (\Delta y_{\alpha} \Delta z_{\alpha}) \quad (10)$$

To find the stationary inertia matrix of a Seaglider with all components (electronics, pumps, wings, rudder, etc.), the vehicle is modeled as a collection of discrete masses. A fixed reference point is selected, which is center of gravity (CG) in this case, and the position of each component is expressed relative to this point (i.e. Δx , Δy , Δz). Each component contributes to the inertia matrix based on its mass and its distance from the CG. All individual contributions from each component are then summed to obtain the total inertia matrix of the system. Each component of the matrix is computed using Equation 5 through Equation 10, where they are summing the inertia contributions from all the components of the seaglider.

This stationary inertia matrix was calculated using the Matlab script `StationaryInertiaCalc.m`.

Step 6.3 — Nominal stationary inertia matrix assembly

$$\mathbf{J}_s = \begin{bmatrix} J_{xx} & J_{xy} & J_{xz} \\ J_{yx} & J_{yy} & J_{yz} \\ J_{zx} & J_{zy} & J_{zz} \end{bmatrix} = \begin{bmatrix} 0.1861 & -0.0007 & 0.1095 \\ -0.0007 & 8.1735 & -0.0000 \\ 0.1095 & -0.0000 & 8.2073 \end{bmatrix} \text{ kg m}^2$$

Fault Cases: Mass and Stationary Inertia Matrices

Case	Component	Mass (g)	x_{CG} (cm)	y_{CG} (cm)	z_{CG} (cm)	Total Mass (g)
Nominal Case						
100% Wing; 100% Rudder	Top Rudder	62.7	170.784	0	10.914	58852
	Starboard Wing	617.8	129.779	29.905	0	
	Bottom Rudder	62.8	170.784	0	-10.914	
	Port Wing	611.3	129.779	-29.905	0	
Fault Cases						
FC1: 50% Starboard Wing; 100% Rudders	Starboard Wing	339.8	128.761	21.08	0	58574
FC2: 25% Starboard Wing; 100% Rudders	Starboard Wing	169.3	128.394	17.456	0	58403.5
FC3: 100% Wings; 0% Top Rudder	Top Rudder	0	0	0	0	58789.2

Table 1: Wing and rudder masses and CG locations for nominal and fault cases from CAD and Trim and Balance sheet (1) (refer to Appendix B on CG locations)

Methodology for calculating component masses for each fault case:

Each mass value is calculated by multiplying the percent area of the flight surface surface area remaining (for the given span-reduced surface) by the total mass of the associated wing or rudder. This assumes wing density is the same throughout the wing. Example: 50% Wing case has 55% of the surface area remaining, therefor 55% of the total mass is remaining. This yields a result of $617.8\text{g} \times 0.55 = 339.8\text{g}$. The percent surface area remaining can be calculated by dividing the 50% wing's surface area by the 100% wing's surface area.

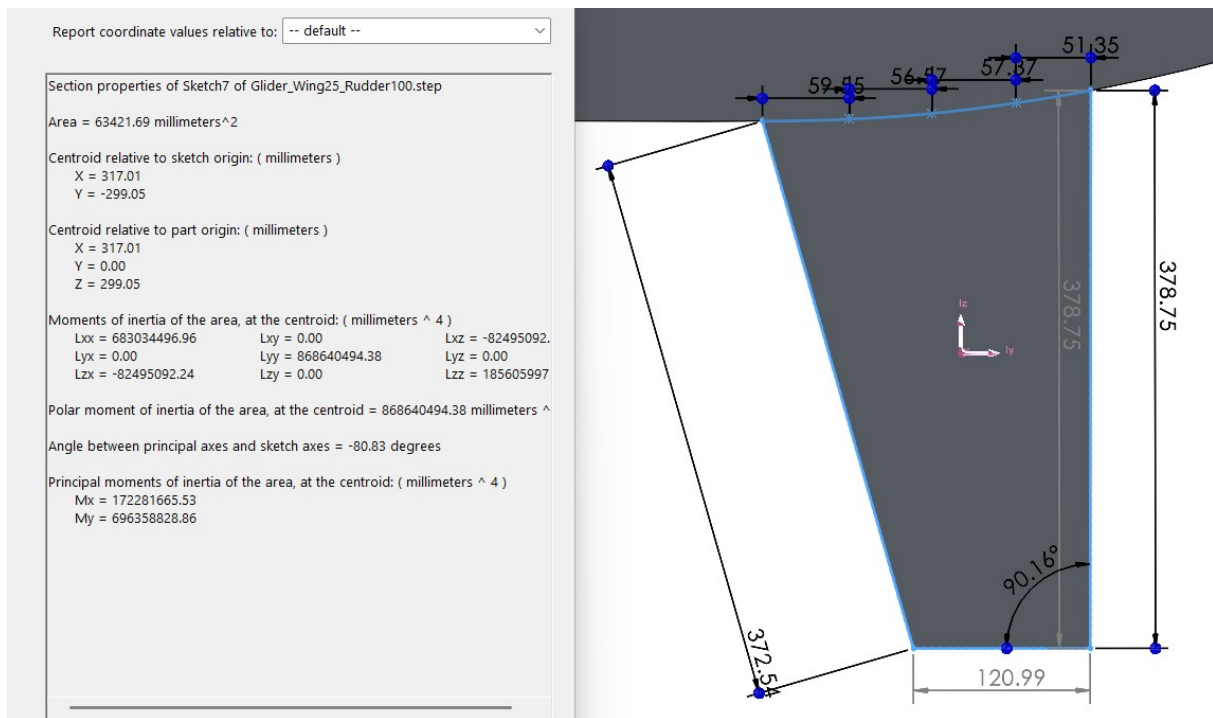


Figure 3: Full wing geometry in SolidWorks. 100% Wing Case Surface Area = 63,421.69 mm^2

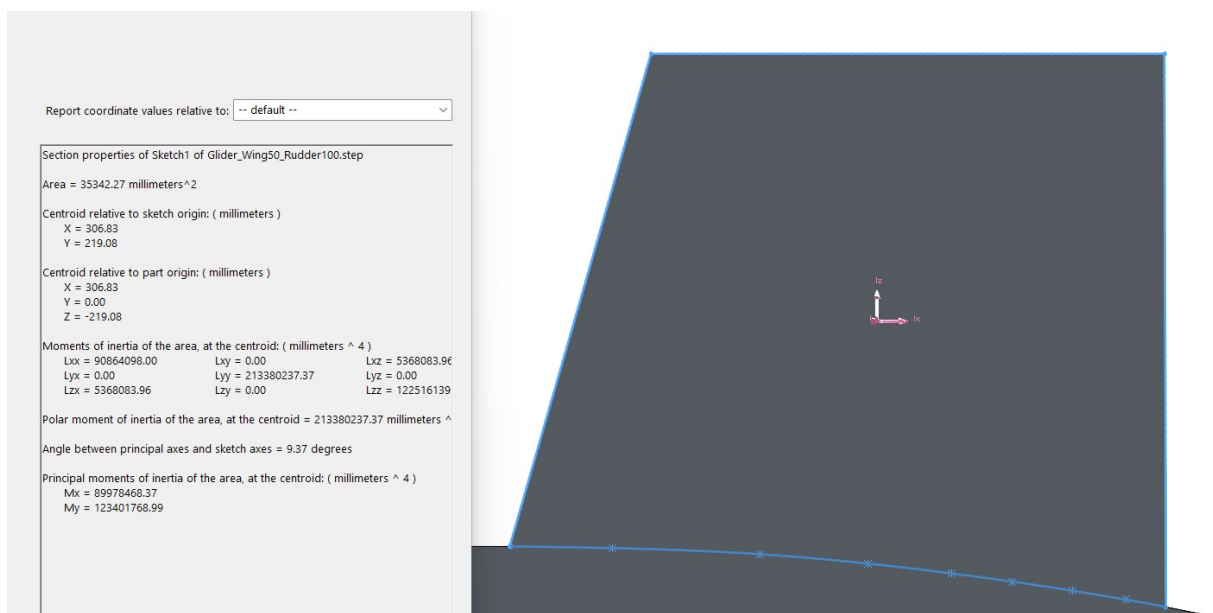


Figure 4: 50% wing geometry in SolidWorks. 50% Wing Case Surface Area = 35,342.27 mm^2

Vehicle CG Shift Analysis due to Faults

The objective of this analysis is to determine how much the center of gravity (CG) of the glider shifts when the Seaglider experiences a wing or rudder loss fault. The two fault cases analyzed are loss of the starboard wing and loss of one rudder since these are the

worst case scenario. The CG shift is evaluated in the x , y , and z directions. In Table 2, x , y , and z distances are measured from nose of the glider and neutral axis of the glider.

Component	Mass (g)	x_{CG} (cm)	y_{CG} (cm)	z_{CG} (cm)	Source
Top Rudder	62.7	170.784	0	10.914	CAD, SG194 trim sheet (1)
Starboard wing	617.8	129.779	29.905	0	CAD, SG194 trim sheet (1)
Total glider	58852	95.752	0	-0.835	SG194 trim sheet (1)

Table 2: Mass and CG Location Data Used for CG Shift Analysis

CG Calculation Method

The nominal CG location of the full glider is calculated using the weighted average of the component masses and their locations:

$$x_{CG} = \frac{\sum m_i x_i}{\sum m_i} \quad (11)$$

When a component is lost, the new CG location is calculated by removing that component's mass contribution from the full-glider CG:

$$x_{CG,\text{fault}} = \frac{M x_{CG} - m_i x_i}{M - m_i} \quad (12)$$

The same method is applied in the y and z directions:

$$y_{CG,\text{fault}} = \frac{M y_{CG} - m_i y_i}{M - m_i} \quad (13)$$

$$z_{CG,\text{fault}} = \frac{M z_{CG} - m_i z_i}{M - m_i} \quad (14)$$

The change in CG location is calculated as:

$$\Delta x_{CG} = |x_{CG,\text{nominal}} - x_{CG,\text{fault}}| \quad (15)$$

$$\Delta y_{CG} = |y_{CG,\text{nominal}} - y_{CG,\text{fault}}| \quad (16)$$

$$\Delta z_{CG} = |z_{CG,\text{nominal}} - z_{CG,\text{fault}}| \quad (17)$$

CG Shift Results

Based on the calculated CG shifts, both component-loss fault cases result in small changes to the center of gravity. The largest shift occurs in the x direction for the loss of the starboard wing, with a CG movement of approximately 3.6 mm. **All calculated CG shifts are less than 5 mm. These are very small shifts relative to the length of the glider which is 1800 mm; therefore, any shifts in CG are neglected (Assumption 5).**

Fault Case	Direction	Nominal CG (cm)	Fault CG (cm)	CG Shift
Loss of starboard wing	x	95.752	95.391	3.60 mm
	y	0	-0.317	3.17 mm
	z	-0.835	-0.844	0.09 mm
Loss of one rudder	x	95.752	95.672	0.80 mm
	y	0	0	0.00 mm
	z	-0.835	-0.8475	0.13 mm

Fault Case	Maximum CG Shift	Result
Loss of starboard wing	3.6 mm	< 5 mm
Loss of one rudder	0.8 mm	< 5 mm

Table 3: Maximum CG Shift for Each Fault Case

Added Mass and Inertia Matrices

Added mass and inertia matrices are the same as the nominal for all fault cases. These matrices only consider the shape and dimensions of the hull and the density of water. Since these properties are not changed due to any of the pre-define fault cases above and we are not considering any damages to the hull of the glider, added mass and inertia matrices are unchanged.

Added mass matrix $\mathbf{M}_f$ (kg)

$$\mathbf{M}_f = \begin{bmatrix} 3.310 & 0 & 0 \\ 0 & 72.005 & 0 \\ 0 & 0 & 72.005 \end{bmatrix}$$

Added inertia matrix $\mathbf{J}_f$ (kg m²)

$$\mathbf{J}_f = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 10.062 & 0 \\ 0 & 0 & 10.062 \end{bmatrix}$$

Stationary Inertia Mass Matrices

Since the wing and rudder masses change due to faults in Table 1, the static inertia mass matrices change. Same steps in Step 6 was followed to obtain these matrices for each fault case. Only change implemented into **Step 6.1** was changing component masses and CG to reflect changes due to fault cases in Table 1. The list of components used in this calculation are in Appendices C, D, and E.

Fault Case	Stationary Inertia Matrix
FC1	$\begin{bmatrix} 0.1459 & 0.0386 & 0.1103 \\ 0.0386 & 8.1390 & 0.0009 \\ 0.1103 & 0.0009 & 8.1326 \end{bmatrix}$
FC2	$\begin{bmatrix} 0.1360 & 0.0526 & 0.1108 \\ 0.0526 & 8.1200 & 0.0013 \\ 0.1108 & 0.0013 & 8.1037 \end{bmatrix}$
FC3	$\begin{bmatrix} 0.1861 & -0.0007 & 0.1095 \\ -0.0007 & 8.1735 & -0.0000 \\ 0.1095 & -0.0000 & 8.2073 \end{bmatrix}$

Table 4: Stationary inertia matrix components for fault cases

Summary

Added mass and added inertia matrix components

Matrix	M_{xx}	M_{yy}	M_{zz}	M_{xy}	M_{xz}	M_{yz}
M_f (kg)	3.310	72.005	72.005	0	0	0
J_f (kg·m ²)	0	10.062	10.062	0	0	0

Stationary inertia components for nominal and fault cases

Case	Total Mass (kg)	J_{xx} (kg·m ²)	J_{yy} (kg·m ²)	J_{zz} (kg·m ²)	J_{xy} (kg·m ²)	J_{xz} (kg·m ²)	J_{yz} (kg·m ²)
Nominal	44.704	0.18607	8.1735	8.2073	-0.0007	0.1095	-1.6231×10^{-5}
50% Wing	44.426	0.14590	8.1390	8.1326	0.03856	0.11031	0.0009
25% Wing	44.255	0.13595	8.1200	8.1037	0.0526	0.1108	0.0013
0% Rudder	44.704	0.18607	8.1735	8.2073	-0.0007	0.10949	-1.6231×10^{-5}

Acknowledgments

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- Holland Kantner provided the Matlab script `StationaryInertiaCalc.m` used to calculate the stationary inertia matrices, and assisted with compiling the data and cleaning it up.
- Letizia Laura assisted with compiling the data, cleaning it up, and implementing it in Matlab.
- Claude AI was only used to prepare the outline (Section flowdown) and all the tables (converting from excel to latex code) in this documentation.

Appendix A: Nominal										
SEAGLIDER - TRIM AND BALANCE		SG 194	Date	24-Oct-13						
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
(8x) 10-32 x 1/2 FHMS, joint ring to aft fairing	55279	1	14.3	0.0143	116.8	0.210	0.00	0.00835	0.000	0
(8x) 10-32 x 1/2 FHMS, Ogive Aft fairing to endcap	55279	1	14.3	0.0143	125.7	0.299	0.00	0.00835	0.000	0
Forward Fairing - with foam	49897	1	4237.1	4.2371	71.7	-0.241	0.00	0.00835	0.000	0
Joint Ring, Fwd to Aft Fairing (bonded to fwd fairing before paint; Long ring = 875g, short ring 688g)	49834	1	688	0.688	116.5	0.207	0.00	0.00835	0.000	0
Nose weight (bonded to fwd fairing after paint)	49844	1	1270.9	1.2709	9.4	-0.864	0.00	0.00835	0.000	0
Nose weight plates	55448	1	495.3	0.4953	9.4	-0.864	0.00	0.00835	0.000	0
(2x) 1/4-20 x 1.5" SHCS + 1/4 LW brass, nose plate to nose base (2 -3 plates)	55185/55253	1	23.4	0.0234	9.4	-0.864	0.00	0.00835	0.000	0
Trim Lead .600" aft of pupa joint ring - bottom (5" strip)		1	695.1	0.6951	106.9	0.111	-13.10	-0.12265	0.000	0
EPDM base (5"x6") + (2x) 2"x 6" tape - bottom		1	24.2	0.0242	106.9	0.111	-12.70	-0.11865	0.000	0
Trim Lead .600" aft of pupa joint ring - port side (5" strip)		1	759.8	0.7598	106.9	0.111	0.00	0.00835	0.000	0
EPDM base (5"x6") + (2x) 2"x 6" tape - port		1	24.0	0.024	106.9	0.111	0.00	0.00835	0.000	0
Trim Lead 600" aft of pupa joint ring - starboard side (5" strip)		1	865.0	0.865	106.9	0.111	0.00	0.00835	0.000	0
EPDM base (5x6") + (2x) 2"x 6" tape - starboard		1	24.0	0.024	106.9	0.111	0.00	0.00835	0.000	0
Trim Lead tape (~1g per 5" lead strip)		15	12.0	0.012	106.9	0.111	13.00	0.13835	0.000	0
(2) Straps - cut to length (start at 34g, subtract trimmed)		1	33.2	0.0332	106.9	0.111	0.00	0.00835	0.000	0
(2) wraps 2x around black elec tape for security		1	12.0	0.012	106.9	0.111	0.00	0.00835	0.000	0
Ogive Aft Fairing - with cutouts for panels	56972	1	2535	2.535	143.2	0.474	-0.10	0.00735	0.000	0
Ogive Wing, Starboard (1m)	49838	1	617.8	0.6178	129.779	0.340	0.00	0.00835	29.905	0.29905
Ogive Wing, Port (1 m)	49838	1	611.3	0.6113	129.779	0.340	0.00	0.00835	-29.905	-0.29905
(8x) 10-32 x 1/2 FHMS, wings to fairing	5xxxx	1	14.3	0.0143	129.6	0.338	0.00	0.00835	0.000	0
Top Rudder			62.7	0.0627	170.784	0.750	10.91	0.10079	0.000	0
Bottom Rudder		1	62.7	0.0627	170.784	0.750	-10.91	-0.10079	0.000	0
(2x) 1/4-20 x 1 1/2" FHMS, rudder to rudder shoe	5xxxx	1	15.7	0.0157	181.8	0.860	0.00	0.00835	0.000	0
Ogive Top panel (with sensor holes)	57651	1	298.1	0.2981	142.8	0.470	11.80	0.12635	0.000	0
Bottom panel with sensor holes	57651	1	290.4	0.2904	142.8	0.470	-11.80	-0.10965	0.000	0
(20x) Ogive Panel screws - 6-32 x 3/8 Truss head MS	56912	1	19.2	0.0192	142.8	0.470	0.00	0.00835	0.000	0
Top panel tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)	NIB	4	14.8	0.0148	132.1	0.363	13.30	0.14135	0.000	0
Bottom panel tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	143	0.472	13.30	0.14135	0.000	0
Aft-end strb Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		2	7.4	0.0074	161.3	0.655	0.00	0.00835	0.000	0
Aft-end port Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		2	7.4	0.0074	161.3	0.655	0.00	0.00835	0.000	0
Under strb-wing Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	132.1	0.363	0.00	0.00835	0.000	0
Under port-wing Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	132.1	0.363	0.00	0.00835	0.000	0
Trim lead to Port & Starbord of puck		1	526	0.526	154.9	0.591	-10.20	-0.09365	0.000	0
Trim lead Aft of puck		1	260.2	0.2602	167.6	0.718	-10.20	-0.09365	0.000	0
Tape for trim lead on bottom panel		3	2.4	0.0024	157.5	0.617	-10.20	-0.09365	0.000	0

Appendix A: Nominal										
SEAGLIDER - TRIM AND BALANCE			SG 194	Date	24-Oct-13					
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Tubes & glue mass on top hatch (4) - non-compressee	NIB	1	71.5	0.0715	154.9	0.591	-13.10	-0.12265	0.000	0
Compressee only on top hatch (4)		1	877.7	0.8777	154.9	0.591	-13.10	-0.12265	0.000	0
Tubes & glue mass on bottom hatch (2)		1	36	0.036	154.9	0.591	-12.70	-0.11865	0.000	0
Compressee only on bottom hatch (2)		1	403.1	0.4031	154.9	0.591	-12.70	-0.11865	0.000	0
Tubes & glue mass under port wing (4)		1	71.4	0.0714	132.1	0.363	0.00	0.00835	0.000	0
Compressee only under port wing (4)		1	876.6	0.8766	132.1	0.363	0.00	0.00835	0.000	0
Tubes & glue mass under starboard wing (4)		1	71.2	0.0712	132.1	0.363	0.00	0.00835	0.000	0
Compressee only under starboard wing (4)		1	896.6	0.8966	132.1	0.363	0.00	0.00835	0.000	0
Tubes & glue mass aft on port side (2)		1	35.6	0.0356	161.3	0.655	0.00	0.00835	0.000	0
Compressee only aft on port side (2)		1	385.8	0.3858	161.3	0.655	0.00	0.00835	0.000	0
Tubes aft & glue mass aft on starboard side (2)		1	35.6	0.0356	161.3	0.655	0.00	0.00835	0.000	0
Compressee only aft on starboard side (2)		1	427.1	0.4271	161.3	0.655	0.00	0.00835	0.000	0
Antenna Assembly GPSI (All parts), including extra comms cable	52336 / 52341 / 52454 / 52476 / 55361	1	628.7	0.6287	211.84	1.161	0.00	0.00835	0.000	0
	52220 / 52485 / 52486 / 55239 / 55321	1	293.0	0.293	142.9	0.471	15.90	0.16735	0.000	0
O2 Sensor (Aanderas optode)	52220 / 52485 / 52486 / 55239 / 55321	1	34.7	0.0347	142.9	0.471	15.90	0.16735	0.000	0
O2 Sensor Assy. Hardware (mount base, SS hose clamp, (2x) 8-32 x 5/8 FHMS, (2x) 8-32 nylok)	55321	1	234.7	0.2347	154.9	0.591	-10.20	-0.09365	0.000	0
Wet Labs Sensor		1								
Wet Labs sensor Assy. Hardware ((2x) snesor clamp, 2(x) 8-32 x 5/8 FHMS - sensor clamp to fairing)	55381 / 52312 / 55245	1	3.2	0.0032	154.9	0.591	-10.20	-0.09365	0.000	0
Position D (SBE-O2): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-2.90	-0.02065	0.000	0
Position E (WL-2): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.10	-0.04265	0.000	0
Position F (WL-3 / PAR): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.80	-0.04965	0.000	0
Position G (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.10	-0.04265	0.000	0
Position H (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-2.90	-0.02065	0.000	0
Position I (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	2.90	0.03735	0.000	0
O-Ring E70-270, Endcap to Batt Hull	52411	1	8.0	0.008	123.3	0.275	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS w/ LW + (5x) 6-32 x 5/8 with zincs, endcap to battery hull	55183 / 55217 / 55214 / 49884	1	13.2	0.0132	123.5	0.277	0.00	0.00835	0.000	0
O-Ring, E70-270, Bulkhead to batt. hull	52411	1	8.0	0.008	72.9	-0.229	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS w/ LW + (5x) 6-32 x 5/8 with zincs, bulkhead to battery hull	55183 / 55217 / 55214 / 49884	1	15.3	0.0153	65.4	-0.304	0.00	0.00835	0.000	0
(4x) 6-32 x 1/2 SHCS + #6 LW, rails to bulkhead	55183 / 55217	1	5.3	0.0053	59.5	-0.363	0.00	0.00835	0.000	0
O-Ring, E70-263, electric hull to bulkhead	52410	1	6.4	0.0064	66.4	-0.294	0.00	0.00835	0.000	0
(8x) 6-32 x 1/2 SHCS + #6 LW, elec hull to bulkhead	55183 / 55217	1	10.5	0.0105	65.8	-0.300	0.00	0.00835	0.000	0
(8x) Ogive fairing adapter clips		1	43.7	0.0437	65.8	-0.300	0.00	0.00835	0.000	0

Appendix A: Nominal										
SEAGLIDER - TRIM AND BALANCE		SG 194	Date	24-Oct-13						
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
(8x) 1/4-28 x 3/8 PHMS, Ogive fairing adapter clip to aft endcap		1	37.1	0.0371	65.8	-0.300	0.00	0.00835	0.000	0
Endcap, Aft	49815	1	2027.6	2.0276	127.0	0.312	0.00	0.00835	0.000	0
(2x) Roll pin, 5/32 OD x 3/8, Pump anti rotation	55248	1	1.0	0.001	123.0	0.272	0.00	0.00835	0.000	0
Connector, Run Tee, 1/8 NPT x 1/4 tube, Push to connect		1	8.4	0.0084	125.2	0.294	0.00	0.00835	0.000	0
Connector, Male, 7/16 straight x 1/4 tube, SS	55506	1	27.6	0.0276	125.2	0.294	1.30	0.02135	0.000	0
Assy, Solenoid Valve, Skinner	52254	1	600.7	0.6007	124.5	0.287	1.50	0.02335	0.000	0
Adaptor, Male, 1/8 NPT x 1/4 M. SS	52430	1	10.5	0.0105	128.1	0.323	1.30	0.02135	0.000	0
Connector, Male, 1/4 tube x 1/8 NPT, Push to connect		1	6.3	0.0063	123.9	0.281	-1.30	-0.00465	0.000	0
(3x) Tubing, 1/4 DIA x 10"		1	12.1	0.0121	119.4	0.236	0.00	0.00835	0.000	0
(2x) Tubing, 1/4 Dia x 4.375"		1	3.5	0.0035	119.4	0.236	0.00	0.00835	0.000	0
Tubing, 1/4 Dia x 1.25"		1	0.5	0.0005	119.4	0.236	0.00	0.00835	0.000	0
Tubing, 1/4 DIA x 8"		1	3.2	0.0032	119.4	0.236	0.00	0.00835	0.000	0
Check Valve, Kepner 204 A-1	52448	1	23.0	0.023	124.7	0.289	-1.30	-0.00465	0.000	0
Branch Tee, Male, 1/4 tube x 1/8 NPT, Push to connect		1	8.4	0.0084	125.2	0.294	0.00	0.00835	0.000	0
Seal Washer	52350	1	0.9	0.0009	129.3	0.335	0.00	0.00835	0.000	0
LeDuc Output Screw	52349	1	7.8	0.0078	129.6	0.338	0.00	0.00835	0.000	0
Hydraulic Pump, LeDuc	52348	1	317.2	0.3172	129.6	0.338	0.00	0.00835	0.000	0
Hydraulic endcap	49803	1	111.1	0.1111	125.9	0.301	0.00	0.00835	0.000	0
Ball Bearing, 8 mm ID x 16 mm OD	52459	1	3.1	0.0031	121.2	0.254	0.00	0.00835	0.000	0
O-Ring, N70-032, hydraulic endcap	52407	1	0.5	0.0005	121.1	0.253	0.00	0.00835	0.000	0
Connector, Male, 1/16 NPT x 1/8 tube, Push to connect		1	5.9	0.0059	122.5	0.267	2.90	0.03735	0.000	0
Tubing, 1/8 Dia. x 10.5"	52418	1	0.9	0.0009	121.3	0.255	0.00	0.00835	0.000	0
Magnetic Seal	52353	1	13.9	0.0139	114.7	0.189	0.00	0.00835	0.000	0
(6x) 6-32 x 1/2 SHCS + #6 LW, hydraulic encap to encap	55183 / 55217	1	7.8	0.0078	121.1	0.253	0.00	0.00835	0.000	0
Lee Plug Set	52445 / 52446	1	0.4	0.0004	122.2	0.264	-1.30	-0.00465	0.000	0
Check Valve (altered item Kepner 1106 A-1-1) + O-Ring (N70 3-906)	49852 / 52408	1	48.1	0.0481	130.8	0.350	0.00	0.00835	0.000	0
Bladder, External	52441	1	837.8	0.8378	133.3	0.375	0.00	0.00835	0.000	0
Nut, Bladder	49802	1	107.3	0.1073	138.7	0.429	0.00	0.00835	0.000	0
Position A (Comms): IE55 w o-ring	52160	1	11.7	0.0117	131.1	0.353	5.10	0.05935	0.000	0
Position B (Opt O2): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	2.90	0.03735	0.000	0
Position C (WL-1): IE55 w o-ring	52160	1	11.5	0.0115	130.8	0.350	0.00	0.00835	0.000	0
Position D (SBE O2): IE55 w o-ring	52160	1	11.8	0.0118	130.8	0.350	-2.90	-0.02065	0.000	0
Position E (WL-2): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.10	-0.04265	0.000	0
Position F (WL-3): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.80	-0.04965	0.000	0
Position G (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.10	-0.04265	0.000	0
Position H (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-2.90	-0.02065	0.000	0
Position I (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	2.90	0.03735	0.000	0

Appendix A: Nominal										
SEAGLIDER - TRIM AND BALANCE		SG 194	Date	24-Oct-13						
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Position J (?): IE55 w o-ring	52160	1	11.8	0.0118	130.8	0.350	5.10	0.05935	0.000	0
Iridium/GPS Ant., conn/o-ring/ CA W5, J2	52241	1	66.1	0.0661	130.8	0.350	0.00	0.00835	0.000	0
Vaccum/pressure relief valve (deep sea power)	52449	1	11.5	0.0115	130.8	0.350	5.80	0.06635	0.000	0
Pump Plate, Inner	49804	1	37.8	0.0378	120.1	0.243	1.90	0.02735	0.000	0
Assy. Main pump motor w/ CA W8, W18	52253	1	130.7	0.1307	123.2	0.274	4.60	0.05435	0.000	0
(3x) M2 x 6 FHMS (motor to interface plate)	55203	1	0.4	0.0004	120.1	0.243	4.60	0.05435	0.000	0
Bore Reducer	52351	1	0.7	0.0007	118.8	0.230	0.00	0.00835	0.000	0
Pulley, LeDuc pump motor	52352	1	5.6	0.0056	118.8	0.230	4.70	0.05535	0.000	0
(6x) 4-40 x 3/8 FHMS, hyd end cap to inner plate	55229	1	1.9	0.0019	120.1	0.243	0.00	0.00835	0.000	0
Pulley, LeDuc pump	49806	1	91.7	0.0917	118.8	0.230	0.00	0.00835	0.000	0
Timing Belt	52357	1	3.4	0.0034	118.8	0.230	1.30	0.02135	0.000	0
Pump Plate, Outer w/ (4x) standoffs	49805	1	20.2	0.0202	118.1	0.223	1.90	0.02735	0.000	0
(4x) 4-40 x 3/16 FHCS		1	1.2	0.0012	118.4	0.226	0.00	0.00835	0.000	0
Check Valve, Kepner 404A-1		1	25.3	0.0253	125.2	0.294	0.00	0.00835	0.000	0
Connector, 45 elbow, 1/8 NPT x 1/4 tube, Push to connect		1	7.2	0.0072	125.2	0.294	0.00	0.00835	0.000	0
Connector, 90 elbow, 1/8 NPT x 1/4 tube, Push to connect		1	7.5	0.0075	125.2	0.294	0.00	0.00835	0.000	0
Connector, Union Tee, 1/4 tube, Push to connect		1	2.9	0.0029	125.2	0.294	0.00	0.00835	0.000	0
Pump Bracket		1	21.5	0.0215	124.2	0.284	0.00	0.00835	0.000	0
Motor, Chrg Pump w/ W34		1	59.9	0.0599	124.2	0.284	0.00	0.00835	0.000	0
(2x) M2 x 6 FHMS, motor to bracket		1	0.2	0.0002	124.2	0.284	2.50	0.03335	0.000	0
Magnet Rotor Housing		1	10.5	0.0105	124.2	0.284	0.00	0.00835	0.000	0
(4x) 6-32 x 1/4 PHMS, bracket to housing		1	1.8	0.0018	124.2	0.284	-1.00	-0.00165	0.000	0
Rotary Magnet		1	18.0	0.018	124.2	0.284	-1.00	-0.00165	0.000	0
Pumphead, Micropump + mounting plate + (3x) 4-40 x 3/8 FHMS		1	224.7	0.2247	124.2	0.284	-2.00	-0.01165	0.000	0
(4x) 4-40 x 3/16 SHCS + #4 LW, housing to flange		1	2.0	0.002	124.2	0.284	-1.30	-0.00465	0.000	0
(2x) Elbow, 1/8 not x 1/4 tube, push to connect		1	15.0	0.015	124.2	0.284	-2.30	-0.01465	0.000	0
Piston, Diaphragm	49811	1	135.3	0.1353	114.3	0.185	0.00	0.00835	0.000	0
Diaphragm, Internal (Bellofram)	52359	1	551.9	0.5519	115.7	0.199	0.00	0.00835	0.000	0
Sikaflex Adhesive Compound	55459	1	15.6	0.0156	114.3	0.185	0.00	0.00835	0.000	0
Cylinder Head, Diaphragm with extra port		1	336.4	0.3364	111.5	0.157	0.00	0.00835	0.000	0
Branch Tee, 1/4 tube x 1/8 NPT, push to connect; 1/4 tube plug		1	9.2	0.0092	109.8	0.140	-6.10	-0.05265	0.000	0
Elbow, 1/8 NPT M x 1/8 NPT M, brass	52426	1	26.9	0.0269	109.8	0.140	6.10	0.06935	0.000	0
Kepner Check Valve 404A-1-200		1	26.6	0.0266	109.8	0.140	2.00	0.02835	0.000	0
90 Elbow, 1/8 npt x 1/8 tube, push to connect		1	7.4	0.0074	109.8	0.140	-0.30	0.00535	0.000	0
Filter, Baston (25 micron) / with zip tie		1	20.9	0.0209	109.8	0.140	2.00	0.02835	0.000	0
Union, 1/4 tube, push to connect		1	1.8	0.0018	109.8	0.140	2.00	0.02835	0.000	0
90 Elbow, 1/8 npt x 1/4 tube, push to connect		1	7.5	0.0075	109.8	0.140	2.00	0.02835	0.000	0

Appendix A: Nominal										
SEAGLIDER - TRIM AND BALANCE		SG 194	Date	24-Oct-13						
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight	Total Weight	Long. CG (x)	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
			grams	kg						
Bleed screw, 10-32 x .125 PHMS with seal	55477	1	1.8	0.0018	109.8	0.140	7.60	0.08435	0.000	0
Cylinder, Diaphragm	49812	1	421.5	0.4215	115.1	0.193	0.00	0.00835	0.000	0
(8x) 5-40 x 3/8 SHCS + #5 LW, head to cylinder	55251	1	7.5	0.0075	110.6	0.148	0.00	0.00835	0.000	0
Standoff (starboard)	55299	1	12.2	0.0122	121.3	0.255	6.60	0.07435	0.000	0
Standoff (port)	55299	1	12.5	0.0125	121.3	0.255	-6.60	-0.05765	0.000	0
(4x) 6-32 x 3/8 SHCS, standoff to cylinder	55183	1	4.5	0.0045	120.7	0.249	0.00	0.00835	0.000	0
(2x) Saddles, mounting, Aft PCB + (4x) 4-40 x 1/4 PHMS	52373 / 55257	1	5.3	0.0053	115.6	0.198	8.10	0.08935	0.000	0
(2x) Saddles, mounting, C PCB + (4x) 4-40 x 1/4 PHMS	49888 / 55257	1	6.7	0.0067	115.6	0.198	3.80	0.04635	0.000	0
(2x) Saddles, mounting, T PCB + (4x) 4-40 x 1/4 PHMS	49889 / 55257	1	6.7	0.0067	115.6	0.198	-3.80	-0.02965	0.000	0
(2x) Saddles, mounting, SBE O2 PCB + (4x) 4-40 x 1/4 SHCS	49890 / 55257	1	4.5	0.0045	115.6	0.198	-8.10	-0.07265	0.000	0
Linear Poteniometer A w/ cable assy (W21), Linear pot clamp, 3-48 x 1/2 SHCS, Spring, Linear pot rod end	52255 / 49814 / 55220	1	40.9	0.0409	121.3	0.255	6.40	0.07235	0.000	0
Linear Potentiometer B w/ cable assy (W20), Linear pot clamp, 3-48 x 1/2 SHCS, Spring, Linear pot rod end	52256 / 49814 / 55220	1	38.5	0.0385	121.3	0.255	6.40	0.07235	0.000	0
(2x) 5-40 x 1/2 SHCS + #5 LW, Clamp to cylinder	55250 / 55212 / 55225	1	1.9	0.0019	121.0	0.252	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS + #6 LW, boost pump bracket to endcap	55183 / 55217	1	2.7	0.0027	122.6	0.268	0.00	0.00835	0.000	0
(4x) 6-32 x 1/2 SHCS + #6 LW, cylinder standoff to endcap	55184 / 55217	1	5.4	0.0054	121.7	0.259	0.00	0.00835	0.000	0
Aft Terminal PCB	55302	1	25.0	0.025	115.6	0.198	8.60	0.09435	0.000	0
SBE Conductivity PCB (included with sensors)	55385	1	32.6	0.0326	115.6	0.198	4.30	0.05135	0.000	0
Conductivity, PCB jumper, CA W26	52261	1	3.1	0.0031	112.8	0.170	3.80	0.04635	0.000	0
Shield, Radiation, C PCB + (4x) 2-56 x 1/4 BHCS	49887 / 55221	1	20.1	0.0201	115.6	0.198	4.80	0.05635	0.000	0
SBE Temperature PCB (included with sensors)	55388	1	22.0	0.022	115.6	0.198	-4.30	-0.03465	0.000	0
Temperature, PCB jumper, CA W25	52260	1	5.2	0.0052	112.8	0.170	1.30	0.02135	0.000	0
Shield, Radiation, T PCB + (4x) 2-56 x 1/4 BHCS	52371 / 55221	1	16.5	0.0165	115.6	0.198	-4.80	-0.03965	0.000	0
Oxygen, PCB jumper, CA W33 (only used with SBE 43f)	52265	1	6.1	0.0061	112.8	0.170	0.00	0.00835	0.000	0
Ribbon Cable (W13) (gray)	52248	1	29.6	0.0296	98.1	0.023	8.90	0.09735	0.000	0
Ribbon Cable (W14) (rainbow)	52249	1	37.1	0.0371	98.1	0.023	8.90	0.09735	0.000	0
CT Sensor Sail Assembly (stanchion, guard, nut, nacelle, (6x) 6-32 x 5/16 FHMS, o-ring (E70-016), o-ring (E70-012), SBE components)	52368	1	293.8	0.2938	121.6	0.258	17.90	0.18735	0.000	0
Mass Shifter Stability Rod Assy w/ Hard Stop ("the stinger")		1	87.6	0.0876	111.5	0.157	0.00	0.00835	0.000	0
Electronic rail, port	49846	1	60.9	0.0609	59.7	-0.361	0.00	0.00835	0.000	0
Electronics rail, starboard	52379	1	60.9	0.0609	59.7	-0.361	0.00	0.00835	0.000	0
Seaglider Mainboard (Mother board, TT8, CF8, flash card)	522990-B.4	1	263.5	0.2635	54.6	-0.412	0.70	0.01535	0.000	0
Insulation Sheet-OEM/Main	52370	1	4.0	0.004	55.2	-0.406	0.44	0.01275	0.000	0
(6x) 6-32 x 1/4 PHMS, main board to rails	55237	1	4.5	0.0045	54.6	-0.412	0.50	0.01335	0.000	0

Appendix A: Nominal

SEAGLIDER - TRIM AND BALANCE

SG 194

Date

24-Oct-13

Vehicle CG

x_CG

95.752

cm

z_CG

-0.835

cm

y_CG

0

cm

Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Assy, OEM Navigation Board (transponder PCB, transformer/inductor, capacitors, GPS reciever, W6 GPS serial interface, GPS antenna cable, DC block, Cable Assembly W7, all soldered on wiring) + RAFOS clock and receiver board & cable to mainboard		1	559.1	0.5591	54.6	-0.412	-0.43	0.00405	0.000	0
(6x) 6-32 x 1/4 PHMS, OEM board to rails	55237	1	4.5	0.0045	54.6	-0.412	-0.43	0.00405	0.000	0
Assy, Iridium Moedm and bracket (includes satellite reciever - iridium LBT, cable assy W3, all clamps for wires, RF switch, DC block, and mounting hardware)	52324	1	271.6	0.2716	54.9	-0.409	1.90	0.02735	0.000	0
4x) 6-32 x 1/4 PHMS + #6 LW, modem cage to rails	55209 / 55217	1	3.4	0.0034	52.8	-0.430	0.29	0.01125	0.000	0
Battery Pack, LV w/ CA W1	52237	1	2664.4	2.6644	54.9	-0.409	-1.81	-0.00975	0.000	0
Battery Pack Cage, LV	49847	1	71.6	0.0716	54.9	-0.409	-1.64	-0.00805	0.000	0
(6x) 6-32 x 1/4 PHMS + #6 LW, batt. cage to rails	55209 / 55217	1	5.4	0.0054	55.3	-0.405	-0.25	0.00585	0.000	0
Hull, Fwd Battery	49817	1	3646.0	3.646	110.6	0.148	0.00	0.00835	0.000	0
Hull, Aft Battery	49817	1	3638.6	3.6386	85.5	-0.103	0.00	0.00835	0.000	0
Joint Ring, battery hulls	49818	1	674.7	0.6747	98.9	0.031	0.00	0.00835	0.000	0
(8x) 6-32 x 3/4 SHCS + #6 LW, joint ring to fwd/aft batt. Hull	55214 / 55217	1	13.3	0.0133	98.1	0.023	0.00	0.00835	0.000	0
(2x) O-Ring, E70-271	52412	1	17.1	0.0171	98.1	0.023	0.00	0.00835	0.000	0
(12x) Clips, Rib	49883	1	25.4	0.0254	98.1	0.023	3.95	0.04785	0.000	0
(12x) Clips, Wiring	52405	1	18.7	0.0187	98.1	0.023	3.95	0.04785	0.000	0
Oil Absorption pad, 1.9" x 10.125" + (2x) rib clips	55527 / 49883	1	11.2	0.0112	98.1	0.023	3.95	0.04785	0.000	0
Endcap, Forward	49830	1	1897.1	1.8971	39.5	-0.563	0.00	0.00835	0.000	0
Acoustic transducer (ITC-3013), w/ cable W36	52267 / 52223	1	1438.7	1.4387	22.9	-0.729	0.00	0.00835	0.000	0
O-Ring, E70-238, transducer to Endcap	52413	1	3.2	0.0032	26.7	-0.691	0.00	0.00835	0.000	0
(6x) Alum 1/4-20 x 3/4 SHCS + 1/4 LW, xducer to end cap	55236 / 55252	1	15.2	0.0152	27.2	-0.686	0.00	0.00835	0.000	0
Hull, Electronics	49829	1	1696.0	1.696	58.0	-0.378	0.00	0.00835	0.000	0
O-Ring, E70-263, elec hull to fwd endcap	52410	1	7.1	0.0071	50.2	-0.456	0.00	0.00835	0.000	0
(8x) 6-32 x 1/2 SHCS + #6 LW, electric hull to fwd encap	55183 / 55217	1	10.3	0.0103	50.5	-0.453	0.00	0.00835	0.000	0
Compass bracket	49848	1	24.4	0.0244	30.5	-0.653	-0.35	0.00485	0.000	0
Compass w/compass carrier PCB	55305 / 52466	1	46.8	0.0468	30.5	-0.653	0.00	0.00835	0.000	0
(4x) 4-40 x 3/8 SHCS + #4 x 1/8L stand off	55256 / 55282	1	2.6	0.0026	30.5	-0.653	-0.31	0.00525	0.000	0
(2x) 4-40 x 3/8 SHCS + #4 LW, comp rnt to fwd encap	55228 / 55233	1	1.3	0.0013	27.4	-0.684	-0.81	0.00025	0.000	0
Cable Assembly W9 (compass cable)	52244	1	21.5	0.0215	38.9	-0.569	0.38	0.01215	0.000	0
Total weight (SUM of components' weight col D)			44704	44.704	Excluding movable battery mass					
Total Scale Weight			58852							

Appendix B

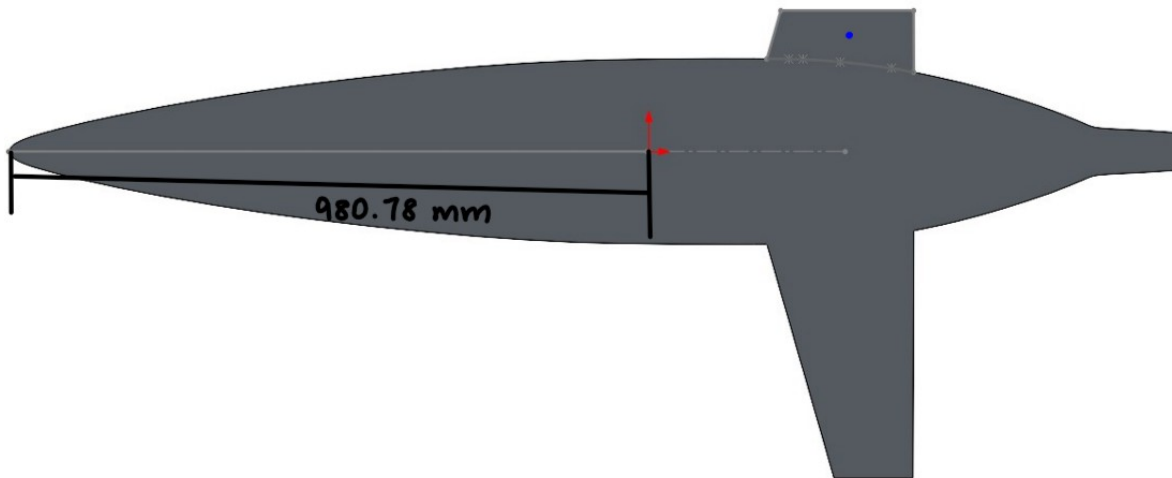


Figure 5: Distance from nose of glider to global coordinate system in x direction

25% wing

$$x_{ca} = 303.16 + 980.78 = 1283.94 \text{ mm} = 128.394 \text{ cm}$$

$$y_{ca} = 174.56 \text{ mm} = 17.456 \text{ mm}$$

$$z_{ca} = 0$$

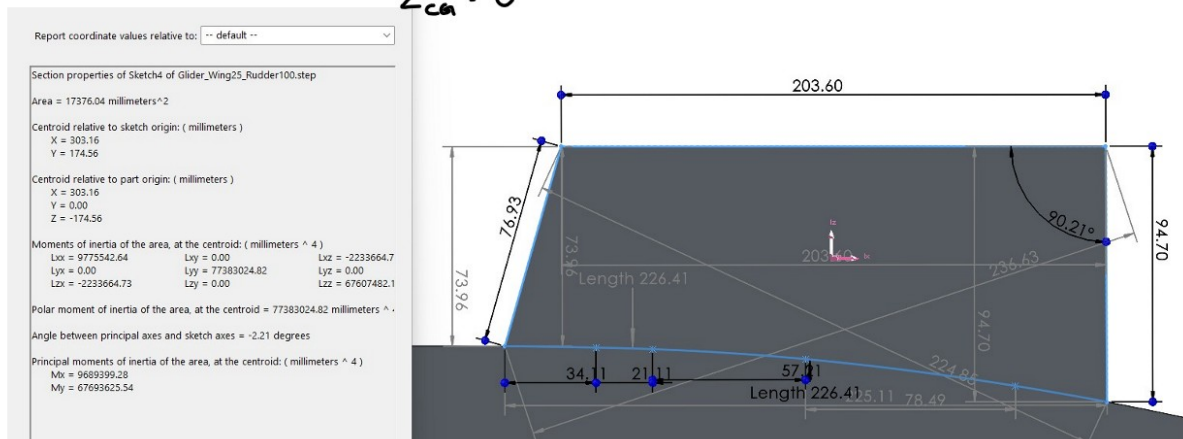


Figure 6: 25% wing remaining

50% wing

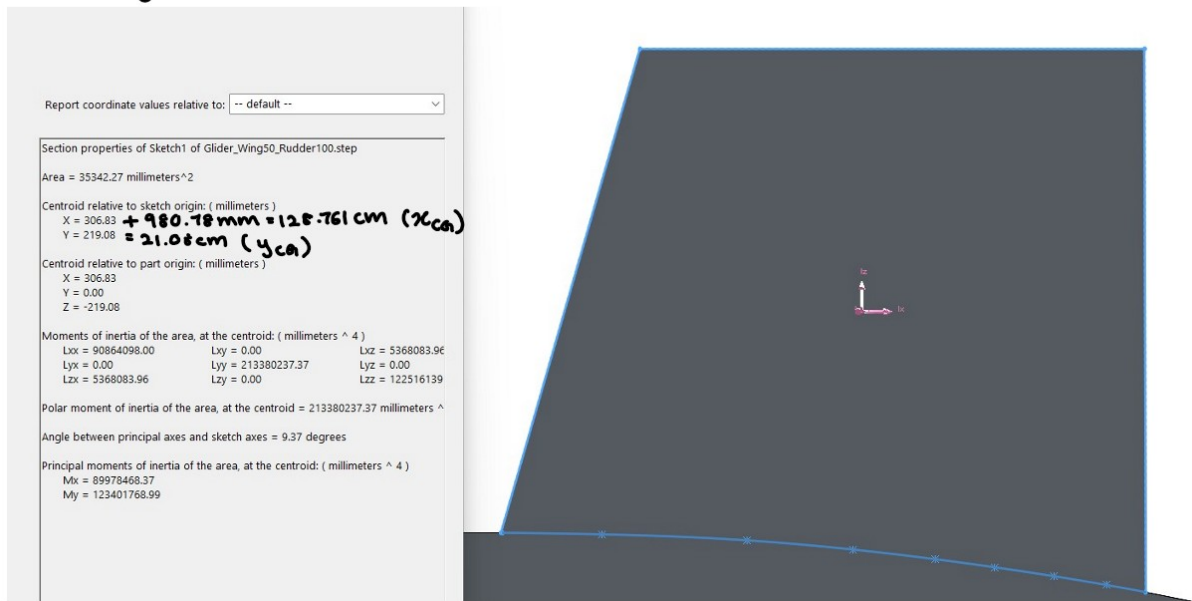


Figure 7: 50% Wing remaining

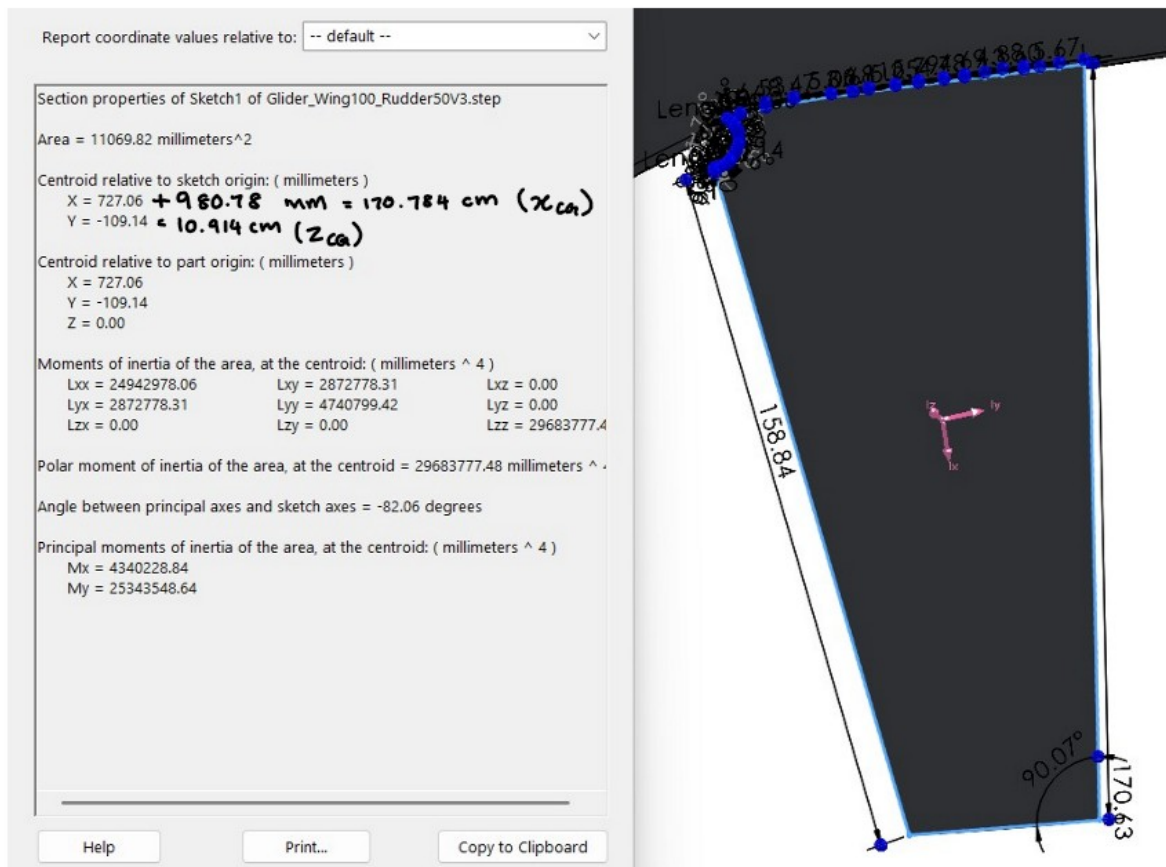


Figure 8: 100% Rudder

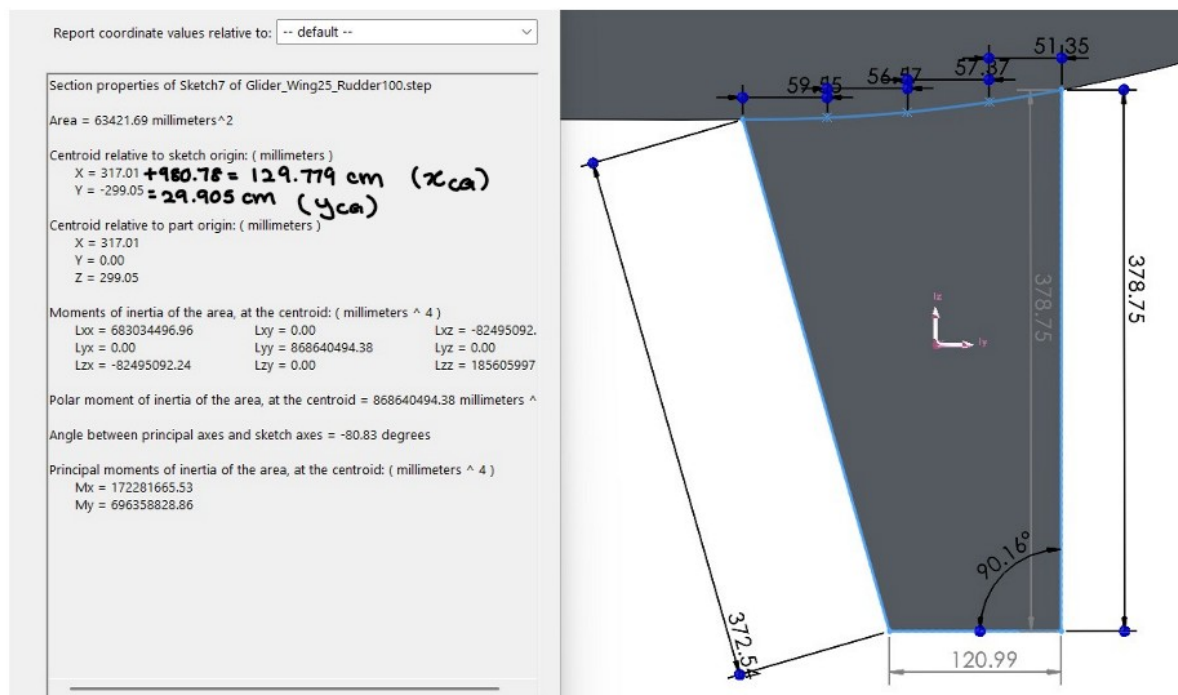


Figure 9: 100% Wing

Appendix C: 50% wing

SEAGLIDER - TRIM AND BALANCE

SG 194

Date

24-Oct-13

Vehicle CG

x_CG

95.752

cm

z_CG

-0.835

cm

y_CG

0

cm

Description	Part #	Qty	Total Weight	Total Weight	Long. CG (x)	Delta_x	VCG&B (z)	Delta_z	Lat. CG (y)	Delta_y
			grams	kg	cm	m	cm	m	cm	m
(8x) 10-32 x 1/2 FHMS, joint ring to aft fairing	55279	1	14.3	0.0143	116.8	0.210	0.00	0.00835	0.000	0
(8x) 10-32 x 1/2 FHMS, Ogive Aft fairing to endcap	55279	1	14.3	0.0143	125.7	0.299	0.00	0.00835	0.000	0
Forward Fairing - with foam	49897	1	4237.1	4.2371	71.7	-0.241	0.00	0.00835	0.000	0
Joint Ring, Fwd to Aft Fairing (bonded to fwd fairing before paint; Long ring = 875g, short ring 688g)	49834	1	688	0.688	116.5	0.207	0.00	0.00835	0.000	0
Nose weight (bonded to fwd fairing after paint)	49844	1	1270.9	1.2709	9.4	-0.864	0.00	0.00835	0.000	0
Nose weight plates	55448	1	495.3	0.4953	9.4	-0.864	0.00	0.00835	0.000	0
(2x) 1/4-20 x 1.5" SHCS + 1/4 LW brass, nose plate to nose base (2 -3 plates)	55185/55253	1	23.4	0.0234	9.4	-0.864	0.00	0.00835	0.000	0
Trim Lead .600" aft of pupa joint ring - bottom (5" strip)		1	695.1	0.6951	106.9	0.111	-13.10	-0.12265	0.000	0
EPDM base (5"x6") + (2x) 2"x 6" tape - bottom		1	24.2	0.0242	106.9	0.111	-12.70	-0.11865	0.000	0
Trim Lead .600" aft of pupa joint ring - port side (5" strip)		1	759.8	0.7598	106.9	0.111	0.00	0.00835	0.000	0
EPDM base (5"x6") + (2x) 2"x 6" tape - port		1	24.0	0.024	106.9	0.111	0.00	0.00835	0.000	0
Trim Lead 600" aft of pupa joint ring - starboard side (5" strip)		1	865.0	0.865	106.9	0.111	0.00	0.00835	0.000	0
EPDM base (5x6") + (2x) 2"x 6" tape - starboard		1	24.0	0.024	106.9	0.111	0.00	0.00835	0.000	0
Trim Lead tape (~1g per 5" lead strip)		15	12.0	0.012	106.9	0.111	13.00	0.13835	0.000	0
(2) Straps - cut to length (start at 34g, subtract trimmed)		1	33.2	0.0332	106.9	0.111	0.00	0.00835	0.000	0
(2) wraps 2x around black elec tape for security		1	12.0	0.012	106.9	0.111	0.00	0.00835	0.000	0
Ogive Aft Fairing - with cutouts for panels	56972	1	2535	2.535	143.2	0.474	-0.10	0.00735	0.000	0
Ogive Wing, Starboard (1m)	49838	1	339.8	0.3398	128.761	0.330	0.00	0.00835	21.080	0.2108
Ogive Wing, Port (1 m)	49838	1	611.3	0.6113	129.779	0.340	0.00	0.00835	-29.905	-0.29905
(8x) 10-32 x 1/2 FHMS, wings to fairing	5xxxx	1	14.3	0.0143	129.6	0.338	0.00	0.00835	0.000	0
Top Rudder		1	62.7	0.0627	170.784	0.750	10.91	0.10079	0.000	0
Bottom Rudder		1	62.7	0.0627	170.784	0.750	-10.91	-0.10079	0.000	0
(2x) 1/4-20 x 1 1/2" FHMS, rudder to rudder shoe	5xxxx	1	15.7	0.0157	181.8	0.860	0.00	0.00835	0.000	0
Ogive Top panel (with sensor holes)	57651	1	298.1	0.2981	142.8	0.470	11.80	0.12635	0.000	0
Bottom panel with sensor holes	57651	1	290.4	0.2904	142.8	0.470	-11.80	-0.10965	0.000	0
(20x) Ogive Panel screws - 6-32 x 3/8 Truss head MS	56912	1	19.2	0.0192	142.8	0.470	0.00	0.00835	0.000	0
Top panel tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)	NIB	4	14.8	0.0148	132.1	0.363	13.30	0.14135	0.000	0
Bottom panel tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	143	0.472	13.30	0.14135	0.000	0
Aft-end strb Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		2	7.4	0.0074	161.3	0.655	0.00	0.00835	0.000	0
Aft-end port Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		2	7.4	0.0074	161.3	0.655	0.00	0.00835	0.000	0
Under strb-wing Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	132.1	0.363	0.00	0.00835	0.000	0
Under port-wing Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	132.1	0.363	0.00	0.00835	0.000	0
Trim lead to Port & Starbord of puck		1	526	0.526	154.9	0.591	-10.20	-0.09365	0.000	0
Trim lead Aft of puck		1	260.2	0.2602	167.6	0.718	-10.20	-0.09365	0.000	0
Tape for trim lead on bottom panel		3	2.4	0.0024	157.5	0.617	-10.20	-0.09365	0.000	0

Appendix C: 50% wing

SEAGLIDER - TRIM AND BALANCE

SG 194

Date

24-Oct-13

Vehicle CG

x_CG

95.752

cm

z_CG

-0.835

cm

y_CG

0

cm

Description	Part #	Qty	Total Weight	Total Weight	Long. CG (x)	Delta_x	VCG&B (z)	Delta_z	Lat. CG (y)	Delta_y
			grams	kg	cm	m	cm	m	cm	m
Tubes & glue mass on top hatch (4) - non-compressee	NIB	1	71.5	0.0715	154.9	0.591	-13.10	-0.12265	0.000	0
Compressee only on top hatch (4)		1	877.7	0.8777	154.9	0.591	-13.10	-0.12265	0.000	0
Tubes & glue mass on bottom hatch (2)		1	36	0.036	154.9	0.591	-12.70	-0.11865	0.000	0
Compressee only on bottom hatch (2)		1	403.1	0.4031	154.9	0.591	-12.70	-0.11865	0.000	0
Tubes & glue mass under port wing (4)		1	71.4	0.0714	132.1	0.363	0.00	0.00835	0.000	0
Compressee only under port wing (4)		1	876.6	0.8766	132.1	0.363	0.00	0.00835	0.000	0
Tubes & glue mass under starboard wing (4)		1	71.2	0.0712	132.1	0.363	0.00	0.00835	0.000	0
Compressee only under starboard wing (4)		1	896.6	0.8966	132.1	0.363	0.00	0.00835	0.000	0
Tubes & glue mass aft on port side (2)		1	35.6	0.0356	161.3	0.655	0.00	0.00835	0.000	0
Compressee only aft on port side (2)		1	385.8	0.3858	161.3	0.655	0.00	0.00835	0.000	0
Tubes aft & glue mass aft on starboard side (2)		1	35.6	0.0356	161.3	0.655	0.00	0.00835	0.000	0
Compressee only aft on starboard side (2)		1	427.1	0.4271	161.3	0.655	0.00	0.00835	0.000	0
Antenna Assembly GPSI (All parts), including extra comms cable	52336 / 52341 / 52454 / 52476 / 55361	1	628.7	0.6287	211.84	1.161	0.00	0.00835	0.000	0
	52220 / 52485 / 52486 / 55239 / 55321	1	293.0	0.293	142.9	0.471	15.90	0.16735	0.000	0
O2 Sensor (Aanderas optode)		1	293.0	0.293	142.9	0.471	15.90	0.16735	0.000	0
O2 Sensor Assy. Hardware (mount base, SS hose clamp, (2x) 8-32 x 5/8 FHMS, (2x) 8-32 nylok)	52220 / 52485 / 52486 / 55239 / 55321	1	34.7	0.0347	142.9	0.471	15.90	0.16735	0.000	0
Wet Labs Sensor		1	234.7	0.2347	154.9	0.591	-10.20	-0.09365	0.000	0
Wet Labs sensor Assy. Hardware ((2x) snesor clamp, 2(x) 8-32 x 5/8 FHMS - sensor clamp to fairing)	55381 / 52312 / 55245	1	3.2	0.0032	154.9	0.591	-10.20	-0.09365	0.000	0
Position D (SBE-O2): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-2.90	-0.02065	0.000	0
Position E (WL-2): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.10	-0.04265	0.000	0
Position F (WL-3 / PAR): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.80	-0.04965	0.000	0
Position G (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.10	-0.04265	0.000	0
Position H (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-2.90	-0.02065	0.000	0
Position I (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	2.90	0.03735	0.000	0
O-Ring E70-270, Endcap to Batt Hull	52411	1	8.0	0.008	123.3	0.275	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS w/ LW + (5x) 6-32 x 5/8 with zincs, endcap to battery hull	55183 / 55217 / 55214 / 49884	1	13.2	0.0132	123.5	0.277	0.00	0.00835	0.000	0
O-Ring, E70-270, Bulkhead to batt. hull	52411	1	8.0	0.008	72.9	-0.229	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS w/ LW + (5x) 6-32 x 5/8 with zincs, bulkhead to battery hull	55183 / 55217 / 55214 / 49884	1	15.3	0.0153	65.4	-0.304	0.00	0.00835	0.000	0
(4x) 6-32 x 1/2 SHCS + #6 LW, rails to bulkhead	55183 / 55217	1	5.3	0.0053	59.5	-0.363	0.00	0.00835	0.000	0
O-Ring, E70-263, electric hull to bulkhead	52410	1	6.4	0.0064	66.4	-0.294	0.00	0.00835	0.000	0
(8x) 6-32 x 1/2 SHCS + #6 LW, elec hull to bulkhead	55183 / 55217	1	10.5	0.0105	65.8	-0.300	0.00	0.00835	0.000	0
(8x) Ogive fairing adapter clips		1	43.7	0.0437	65.8	-0.300	0.00	0.00835	0.000	0

Appendix C: 50% wing											
SEAGLIDER - TRIM AND BALANCE			SG 194	Date	24-Oct-13						
	Vehicle CG	x_CG	95.752		cm						
		z_CG	-0.835		cm						
		y_CG	0		cm						
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m	
(8x) 1/4-28 x 3/8 PHMS, Ogive fairing adapter clip to aft endcap		1	37.1	0.0371	65.8	-0.300	0.00	0.00835	0.000	0	
Endcap, Aft	49815	1	2027.6	2.0276	127.0	0.312	0.00	0.00835	0.000	0	
(2x) Roll pin, 5/32 OD x 3/8, Pump anti rotation	55248	1	1.0	0.001	123.0	0.272	0.00	0.00835	0.000	0	
Connector, Run Tee, 1/8 NPT x 1/4 tube, Push to connect		1	8.4	0.0084	125.2	0.294	0.00	0.00835	0.000	0	
Connector, Male, 7/16 straight x 1/4 tube, SS	55506	1	27.6	0.0276	125.2	0.294	1.30	0.02135	0.000	0	
Assy, Solenoid Valve, Skinner	52254	1	600.7	0.6007	124.5	0.287	1.50	0.02335	0.000	0	
Adaptor, Male, 1/8 NPT x 1/4 M, SS	52430	1	10.5	0.0105	128.1	0.323	1.30	0.02135	0.000	0	
Connector, Male, 1/4 tube x 1/8 NPT, Push to connect		1	6.3	0.0063	123.9	0.281	-1.30	-0.00465	0.000	0	
(3x) Tubing, 1/4 DIA x 10"		1	12.1	0.0121	119.4	0.236	0.00	0.00835	0.000	0	
(2x) Tubing, 1/4 Dia x 4.375"		1	3.5	0.0035	119.4	0.236	0.00	0.00835	0.000	0	
Tubing, 1/4 Dia x 1.25"		1	0.5	0.0005	119.4	0.236	0.00	0.00835	0.000	0	
Tubing, 1/4 DIA x 8"		1	3.2	0.0032	119.4	0.236	0.00	0.00835	0.000	0	
Check Valve, Kepner 204 A-1	52448	1	23.0	0.023	124.7	0.289	-1.30	-0.00465	0.000	0	
Branch Tee, Male, 1/4 tube x 1/8 NPT, Push to connect		1	8.4	0.0084	125.2	0.294	0.00	0.00835	0.000	0	
Seal Washer	52350	1	0.9	0.0009	129.3	0.335	0.00	0.00835	0.000	0	
LeDuc Output Screw	52349	1	7.8	0.0078	129.6	0.338	0.00	0.00835	0.000	0	
Hydraulic Pump, LeDuc	52348	1	317.2	0.3172	129.6	0.338	0.00	0.00835	0.000	0	
Hydraulic endcap	49803	1	111.1	0.1111	125.9	0.301	0.00	0.00835	0.000	0	
Ball Bearing, 8 mm ID x 16 mm OD	52459	1	3.1	0.0031	121.2	0.254	0.00	0.00835	0.000	0	
O-Ring, N70-032, hydraulic endcap	52407	1	0.5	0.0005	121.1	0.253	0.00	0.00835	0.000	0	
Connector, Male, 1/16 NPT x 1/8 tube, Push to connect		1	5.9	0.0059	122.5	0.267	2.90	0.03735	0.000	0	
Tubing, 1/8 Dia. x 10.5"	52418	1	0.9	0.0009	121.3	0.255	0.00	0.00835	0.000	0	
Magnetic Seal	52353	1	13.9	0.0139	114.7	0.189	0.00	0.00835	0.000	0	
(6x) 6-32 x 1/2 SHCS + #6 LW, hydraulic encap to encap	55183 / 55217	1	7.8	0.0078	121.1	0.253	0.00	0.00835	0.000	0	
Lee Plug Set	52445 / 52446	1	0.4	0.0004	122.2	0.264	-1.30	-0.00465	0.000	0	
Check Valve (altered item Kepner 1106 A-1-1) + O-Ring (N70 3-906)	49852 / 52408	1	48.1	0.0481	130.8	0.350	0.00	0.00835	0.000	0	
Bladder, External	52441	1	837.8	0.8378	133.3	0.375	0.00	0.00835	0.000	0	
Nut, Bladder	49802	1	107.3	0.1073	138.7	0.429	0.00	0.00835	0.000	0	
Position A (Comms): IE55 w o-ring	52160	1	11.7	0.0117	131.1	0.353	5.10	0.05935	0.000	0	
Position B (Opt O2): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	2.90	0.03735	0.000	0	
Position C (WL-1): IE55 w o-ring	52160	1	11.5	0.0115	130.8	0.350	0.00	0.00835	0.000	0	
Position D (SBE O2): IE55 w o-ring	52160	1	11.8	0.0118	130.8	0.350	-2.90	-0.02065	0.000	0	
Position E (WL-2): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.10	-0.04265	0.000	0	
Position F (WL-3): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.80	-0.04965	0.000	0	
Position G (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.10	-0.04265	0.000	0	
Position H (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-2.90	-0.02065	0.000	0	
Position I (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	2.90	0.03735	0.000	0	

Appendix C: 50% wing										
SEAGLIDER - TRIM AND BALANCE			SG 194	Date	24-Oct-13					
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Position J (?): IE55 w o-ring	52160	1	11.8	0.0118	130.8	0.350	5.10	0.05935	0.000	0
Iridium/GPS Ant., conn/o-ring/ CA W5, J2	52241	1	66.1	0.0661	130.8	0.350	0.00	0.00835	0.000	0
Vaccum/pressure relief valve (deep sea power)	52449	1	11.5	0.0115	130.8	0.350	5.80	0.06635	0.000	0
Pump Plate, Inner	49804	1	37.8	0.0378	120.1	0.243	1.90	0.02735	0.000	0
Assy. Main pump motor w/ CA W8, W18	52253	1	130.7	0.1307	123.2	0.274	4.60	0.05435	0.000	0
(3x) M2 x 6 FHMS (motor to interface plate)	55203	1	0.4	0.0004	120.1	0.243	4.60	0.05435	0.000	0
Bore Reducer	52351	1	0.7	0.0007	118.8	0.230	0.00	0.00835	0.000	0
Pulley, LeDuc pump motor	52352	1	5.6	0.0056	118.8	0.230	4.70	0.05535	0.000	0
(6x) 4-40 x 3/8 FHMS, hyd end cap to inner plate	55229	1	1.9	0.0019	120.1	0.243	0.00	0.00835	0.000	0
Pulley, LeDuc pump	49806	1	91.7	0.0917	118.8	0.230	0.00	0.00835	0.000	0
Timing Belt	52357	1	3.4	0.0034	118.8	0.230	1.30	0.02135	0.000	0
Pump Plate, Outer w/ (4x) standoffs	49805	1	20.2	0.0202	118.1	0.223	1.90	0.02735	0.000	0
(4x) 4-40 x 3/16 FHCS		1	1.2	0.0012	118.4	0.226	0.00	0.00835	0.000	0
Check Valve, Kepner 404A-1		1	25.3	0.0253	125.2	0.294	0.00	0.00835	0.000	0
Connector, 45 elbow, 1/8 NPT x 1/4 tube, Push to connect		1	7.2	0.0072	125.2	0.294	0.00	0.00835	0.000	0
Connector, 90 elbow, 1/8 NPT x 1/4 tube, Push to connect		1	7.5	0.0075	125.2	0.294	0.00	0.00835	0.000	0
Connector, Union Tee, 1/4 tube, Push to connect		1	2.9	0.0029	125.2	0.294	0.00	0.00835	0.000	0
Pump Bracket		1	21.5	0.0215	124.2	0.284	0.00	0.00835	0.000	0
Motor, Chrg Pump w/ W34		1	59.9	0.0599	124.2	0.284	0.00	0.00835	0.000	0
(2x) M2 x 6 FHMS, motor to bracket		1	0.2	0.0002	124.2	0.284	2.50	0.03335	0.000	0
Magnet Rotor Housing		1	10.5	0.0105	124.2	0.284	0.00	0.00835	0.000	0
(4x) 6-32 x 1/4 PHMS, bracket to housing		1	1.8	0.0018	124.2	0.284	-1.00	-0.00165	0.000	0
Rotary Magnet		1	18.0	0.018	124.2	0.284	-1.00	-0.00165	0.000	0
Pumphead, Micropump + mounting plate + (3x) 4-40 x 3/8 FHMS		1	224.7	0.2247	124.2	0.284	-2.00	-0.01165	0.000	0
(4x) 4-40 x 3/16 SHCS + #4 LW, housing to flange		1	2.0	0.002	124.2	0.284	-1.30	-0.00465	0.000	0
(2x) Elbow, 1/8 not x 1/4 tube, push to connect		1	15.0	0.015	124.2	0.284	-2.30	-0.01465	0.000	0
Piston, Diaphragm	49811	1	135.3	0.1353	114.3	0.185	0.00	0.00835	0.000	0
Diaphragm, Internal (Bellofram)	52359	1	551.9	0.5519	115.7	0.199	0.00	0.00835	0.000	0
Sikaflex Adhesive Compound	55459	1	15.6	0.0156	114.3	0.185	0.00	0.00835	0.000	0
Cylinder Head, Diaphragm with extra port		1	336.4	0.3364	111.5	0.157	0.00	0.00835	0.000	0
Branch Tee, 1/4 tube x 1/8 NPT, push to connect; 1/4 tube plug		1	9.2	0.0092	109.8	0.140	-6.10	-0.05265	0.000	0
Elbow, 1/8 NPT M x 1/8 NPT M, brass	52426	1	26.9	0.0269	109.8	0.140	6.10	0.06935	0.000	0
Kepner Check Valve 404A-1-200		1	26.6	0.0266	109.8	0.140	2.00	0.02835	0.000	0
90 Elbow, 1/8 npt x 1/8 tube, push to connect		1	7.4	0.0074	109.8	0.140	-0.30	0.00535	0.000	0
Filter, Baston (25 micron) / with zip tie		1	20.9	0.0209	109.8	0.140	2.00	0.02835	0.000	0
Union, 1/4 tube, push to connect		1	1.8	0.0018	109.8	0.140	2.00	0.02835	0.000	0
90 Elbow, 1/8 npt x 1/4 tube, push to connect		1	7.5	0.0075	109.8	0.140	2.00	0.02835	0.000	0

Appendix C: 50% wing										
SEAGLIDER - TRIM AND BALANCE			SG 194	Date	24-Oct-13					
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight	Total Weight	Long. CG (x)	Delta_x	VCG&B (z)	Delta_z	Lat. CG (y)	Delta_y
			grams	kg	cm	m	cm	m	cm	m
Bleed screw, 10-32 x .125 PHMS with seal	55477	1	1.8	0.0018	109.8	0.140	7.60	0.08435	0.000	0
Cylinder, Diaphragm	49812	1	421.5	0.4215	115.1	0.193	0.00	0.00835	0.000	0
(8x) 5-40 x 3/8 SHCS + #5 LW, head to cylinder	55251	1	7.5	0.0075	110.6	0.148	0.00	0.00835	0.000	0
Standoff (starboard)	55299	1	12.2	0.0122	121.3	0.255	6.60	0.07435	0.000	0
Standoff (port)	55299	1	12.5	0.0125	121.3	0.255	-6.60	-0.05765	0.000	0
(4x) 6-32 x 3/8 SHCS, standoff to cylinder	55183	1	4.5	0.0045	120.7	0.249	0.00	0.00835	0.000	0
(2x) Saddles, mounting, Aft PCB + (4x) 4-40 x 1/4 PHMS	52373 / 55257	1	5.3	0.0053	115.6	0.198	8.10	0.08935	0.000	0
(2x) Saddles, mounting, C PCB + (4x) 4-40 x 1/4 PHMS	49888 / 55257	1	6.7	0.0067	115.6	0.198	3.80	0.04635	0.000	0
(2x) Saddles, mounting, T PCB + (4x) 4-40 x 1/4 PHMS	49889 / 55257	1	6.7	0.0067	115.6	0.198	-3.80	-0.02965	0.000	0
(2x) Saddles, mounting, SBE O2 PCB + (4x) 4-40 x 1/4 SHCS	49890 / 55257	1	4.5	0.0045	115.6	0.198	-8.10	-0.07265	0.000	0
Linear Poteniometer A w/ cable assy (W21), Linear pot clamp, 3-48 x 1/2 SHCS, Spring, Linear pot rod end	52255 / 49814 / 55220	1	40.9	0.0409	121.3	0.255	6.40	0.07235	0.000	0
Linear Potentiometer B w/ cable assy (W20), Linear pot clamp, 3-48 x 1/2 SHCS, Spring, Linear pot rod end	52256 / 49814 / 55220	1	38.5	0.0385	121.3	0.255	6.40	0.07235	0.000	0
(2x) 5-40 x 1/2 SHCS + #5 LW, Clamp to cylinder	55250 / 55212 / 55225	1	1.9	0.0019	121.0	0.252	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS + #6 LW, boost pump bracket to endcap	55183 / 55217	1	2.7	0.0027	122.6	0.268	0.00	0.00835	0.000	0
(4x) 6-32 x 1/2 SHCS + #6 LW, cylinder standoff to endcap	55184 / 55217	1	5.4	0.0054	121.7	0.259	0.00	0.00835	0.000	0
Aft Terminal PCB	55302	1	25.0	0.025	115.6	0.198	8.60	0.09435	0.000	0
SBE Conductivity PCB (included with sensors)	55385	1	32.6	0.0326	115.6	0.198	4.30	0.05135	0.000	0
Conductivity, PCB jumper, CA W26	52261	1	3.1	0.0031	112.8	0.170	3.80	0.04635	0.000	0
Shield, Radiation, C PCB + (4x) 2-56 x 1/4 BHCS	49887 / 55221	1	20.1	0.0201	115.6	0.198	4.80	0.05635	0.000	0
SBE Temperature PCB (included with sensors)	55388	1	22.0	0.022	115.6	0.198	-4.30	-0.03465	0.000	0
Temperature, PCB jumper, CA W25	52260	1	5.2	0.0052	112.8	0.170	1.30	0.02135	0.000	0
Shield, Radiation, T PCB + (4x) 2-56 x 1/4 BHCS	52371 / 55221	1	16.5	0.0165	115.6	0.198	-4.80	-0.03965	0.000	0
Oxygen, PCB jumper, CA W33 (only used with SBE 43f)	52265	1	6.1	0.0061	112.8	0.170	0.00	0.00835	0.000	0
Ribbon Cable (W13) (gray)	52248	1	29.6	0.0296	98.1	0.023	8.90	0.09735	0.000	0
Ribbon Cable (W14) (rainbow)	52249	1	37.1	0.0371	98.1	0.023	8.90	0.09735	0.000	0
CT Sensor Sail Assembly (stanchion, guard, nut, nacelle, (6x) 6-32 x 5/16 FHMS, o-ring (E70-016), o-ring (E70-012), SBE components)	52368	1	293.8	0.2938	121.6	0.258	17.90	0.18735	0.000	0
Mass Shifter Stability Rod Assy w/ Hard Stop ("the stinger")		1	87.6	0.0876	111.5	0.157	0.00	0.00835	0.000	0
Electronic rail, port	49846	1	60.9	0.0609	59.7	-0.361	0.00	0.00835	0.000	0
Electronics rail, starboard	52379	1	60.9	0.0609	59.7	-0.361	0.00	0.00835	0.000	0
Seaglider Mainboard (Mother board, TT8, CF8, flash card)	522990-B.4	1	263.5	0.2635	54.6	-0.412	0.70	0.01535	0.000	0
Insulation Sheet-OEM/Main	52370	1	4.0	0.004	55.2	-0.406	0.44	0.01275	0.000	0
(6x) 6-32 x 1/4 PHMS, main board to rails	55237	1	4.5	0.0045	54.6	-0.412	0.50	0.01335	0.000	0

Appendix C: 50% wing

SEAGLIDER - TRIM AND BALANCE	SG 194	Date	24-Oct-13							
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Assy, OEM Navigation Board (transponder PCB, transformer/inductor, capacitors, GPS reciever, W6 GPS serial interface, GPS antenna cable, DC block, Cable Assembly W7, all soldered on wiring) + RAFOS clock and receiver board & cable to mainboard		1	559.1	0.5591	54.6	-0.412	-0.43	0.00405	0.000	0
(6x) 6-32 x 1/4 PHMS, OEM board to rails	55237	1	4.5	0.0045	54.6	-0.412	-0.43	0.00405	0.000	0
Assy, Iridium Moedm and bracket (includes satellite reciever - iridium LBT, cable assy W3, all clamps for wires, RF switch, DC block, and mounting hardware)	52324	1	271.6	0.2716	54.9	-0.409	1.90	0.02735	0.000	0
4x) 6-32 x 1/4 PHMS + #6 LW, modem cage to rails	55209 / 55217	1	3.4	0.0034	52.8	-0.430	0.29	0.01125	0.000	0
Battery Pack, LV w/ CA W1	52237	1	2664.4	2.6644	54.9	-0.409	-1.81	-0.00975	0.000	0
Battery Pack Cage, LV	49847	1	71.6	0.0716	54.9	-0.409	-1.64	-0.00805	0.000	0
(6x) 6-32 x 1/4 PHMS + #6 LW, batt. cage to rails	55209 / 55217	1	5.4	0.0054	55.3	-0.405	-0.25	0.00585	0.000	0
Hull, Fwd Battery	49817	1	3646.0	3.646	110.6	0.148	0.00	0.00835	0.000	0
Hull, Aft Battery	49817	1	3638.6	3.6386	85.5	-0.103	0.00	0.00835	0.000	0
Joint Ring, battery hulls	49818	1	674.7	0.6747	98.9	0.031	0.00	0.00835	0.000	0
(8x) 6-32 x 3/4 SHCS + #6 LW, joint ring to fwd/aft batt. Hull	55214 / 55217	1	13.3	0.0133	98.1	0.023	0.00	0.00835	0.000	0
(2x) O-Ring, E70-271	52412	1	17.1	0.0171	98.1	0.023	0.00	0.00835	0.000	0
(12x) Clips, Rib	49883	1	25.4	0.0254	98.1	0.023	3.95	0.04785	0.000	0
(12x) Clips, Wiring	52405	1	18.7	0.0187	98.1	0.023	3.95	0.04785	0.000	0
Oil Absorption pad, 1.9" x 10.125" + (2x) rib clips	55527 / 49883	1	11.2	0.0112	98.1	0.023	3.95	0.04785	0.000	0
Endcap, Forward	49830	1	1897.1	1.8971	39.5	-0.563	0.00	0.00835	0.000	0
Acoustic transducer (ITC-3013), w/ cable W36	52267 / 52223	1	1438.7	1.4387	22.9	-0.729	0.00	0.00835	0.000	0
O-Ring, E70-238, transducer to Endcap	52413	1	3.2	0.0032	26.7	-0.691	0.00	0.00835	0.000	0
(6x) Alum 1/4-20 x 3/4 SHCS + 1/4 LW, xducer to end cap	55236 / 55252	1	15.2	0.0152	27.2	-0.686	0.00	0.00835	0.000	0
Hull, Electronics	49829	1	1696.0	1.696	58.0	-0.378	0.00	0.00835	0.000	0
O-Ring, E70-263, elec hull to fwd endcap	52410	1	7.1	0.0071	50.2	-0.456	0.00	0.00835	0.000	0
(8x) 6-32 x 1/2 SHCS + #6 LW, electric hull to fwd encap	55183 / 55217	1	10.3	0.0103	50.5	-0.453	0.00	0.00835	0.000	0
Compass bracket	49848	1	24.4	0.0244	30.5	-0.653	-0.35	0.00485	0.000	0
Compass w/compass carrier PCB	55305 / 52466	1	46.8	0.0468	30.5	-0.653	0.00	0.00835	0.000	0
(4x) 4-40 x 3/8 SHCS + #4 x 1/8L stand off	55256 / 55282	1	2.6	0.0026	30.5	-0.653	-0.31	0.00525	0.000	0
(2x) 4-40 x 3/8 SHCS + #4 LW, comp mt to fwd encap	55228 / 55233	1	1.3	0.0013	27.4	-0.684	-0.81	0.00025	0.000	0
Cable Assembly W9 (compass cable)	52244	1	21.5	0.0215	38.9	-0.569	0.38	0.01215	0.000	0
Total weight (SUM of components' weight col D)			44426	44.426	Excluding movable battery mass					
Total Scale Weight			58852							

Appendix D: 25% wing

SEAGLIDER - TRIM AND BALANCE

SG 194

Date

24-Oct-13

Vehicle CG

x_CG

95.752

cm

z_CG

-0.835

cm

y_CG

0

cm

Description	Part #	Qty	Total Weight	Total Weight	Long. CG (x)	Delta_x	VCG&B (z)	Delta_z	Lat. CG (y)	Delta_y
			grams	kg	cm	m	cm	m	cm	m
(8x) 10-32 x 1/2 FHMS, joint ring to aft fairing	55279	1	14.3	0.0143	116.8	0.210	0.00	0.00835	0.000	0
(8x) 10-32 x 1/2 FHMS, Ogive Aft fairing to endcap	55279	1	14.3	0.0143	125.7	0.299	0.00	0.00835	0.000	0
Forward Fairing - with foam	49897	1	4237.1	4.2371	71.7	-0.241	0.00	0.00835	0.000	0
Joint Ring, Fwd to Aft Fairing (bonded to fwd fairing before paint; Long ring = 875g, short ring 688g)	49834	1	688	0.688	116.5	0.207	0.00	0.00835	0.000	0
Nose weight (bonded to fwd fairing after paint)	49844	1	1270.9	1.2709	9.4	-0.864	0.00	0.00835	0.000	0
Nose weight plates	55448	1	495.3	0.4953	9.4	-0.864	0.00	0.00835	0.000	0
(2x) 1/4-20 x 1.5" SHCS + 1/4 LW brass, nose plate to nose base (2 -3 plates)	55185/55253	1	23.4	0.0234	9.4	-0.864	0.00	0.00835	0.000	0
Trim Lead .600" aft of pupa joint ring - bottom (5" strip)		1	695.1	0.6951	106.9	0.111	-13.10	-0.12265	0.000	0
EPDM base (5"x6") + (2x) 2"x 6" tape - bottom		1	24.2	0.0242	106.9	0.111	-12.70	-0.11865	0.000	0
Trim Lead .600" aft of pupa joint ring - port side (5" strip)		1	759.8	0.7598	106.9	0.111	0.00	0.00835	0.000	0
EPDM base (5"x6") + (2x) 2"x 6" tape - port		1	24.0	0.024	106.9	0.111	0.00	0.00835	0.000	0
Trim Lead 600" aft of pupa joint ring - starboard side (5" strip)		1	865.0	0.865	106.9	0.111	0.00	0.00835	0.000	0
EPDM base (5x6") + (2x) 2"x 6" tape - starboard		1	24.0	0.024	106.9	0.111	0.00	0.00835	0.000	0
Trim Lead tape (~1g per 5" lead strip)		15	12.0	0.012	106.9	0.111	13.00	0.13835	0.000	0
(2) Straps - cut to length (start at 34g, subtract trimmed)		1	33.2	0.0332	106.9	0.111	0.00	0.00835	0.000	0
(2) wraps 2x around black elec tape for security		1	12.0	0.012	106.9	0.111	0.00	0.00835	0.000	0
Ogive Aft Fairing - with cutouts for panels	56972	1	2535	2.535	143.2	0.474	-0.10	0.00735	0.000	0
Ogive Wing, Starboard (1m)	49838	1	169.3	0.1693	128.394	0.326	0.00	0.00835	17.456	0.17456
Ogive Wing, Port (1 m)	49838	1	611.3	0.6113	129.779	0.340	0.00	0.00835	-29.905	-0.29905
(8x) 10-32 x 1/2 FHMS, wings to fairing	5xxxx	1	14.3	0.0143	129.6	0.338	0.00	0.00835	0.000	0
Top Rudder		1	62.7	0.0627	170.784	0.750	10.91	0.10079	0.000	0
Bottom Rudder		1	62.7	0.0627	170.784	0.750	-10.91	-0.10079	0.000	0
(2x) 1/4-20 x 1 1/2" FHMS, rudder to rudder shoe	5xxxx	1	15.7	0.0157	181.8	0.860	0.00	0.00835	0.000	0
Ogive Top panel (with sensor holes)	57651	1	298.1	0.2981	142.8	0.470	11.80	0.12635	0.000	0
Bottom panel with sensor holes	57651	1	290.4	0.2904	142.8	0.470	-11.80	-0.10965	0.000	0
(20x) Ogive Panel screws - 6-32 x 3/8 Truss head MS	56912	1	19.2	0.0192	142.8	0.470	0.00	0.00835	0.000	0
Top panel tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)	NIB	4	14.8	0.0148	132.1	0.363	13.30	0.14135	0.000	0
Bottom panel tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	143	0.472	13.30	0.14135	0.000	0
Aft-end strb Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		2	7.4	0.0074	161.3	0.655	0.00	0.00835	0.000	0
Aft-end port Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		2	7.4	0.0074	161.3	0.655	0.00	0.00835	0.000	0
Under strb-wing Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	132.1	0.363	0.00	0.00835	0.000	0
Under port-wing Tie-downs for compressee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	132.1	0.363	0.00	0.00835	0.000	0
Trim lead to Port & Starbord of puck		1	526	0.526	154.9	0.591	-10.20	-0.09365	0.000	0
Trim lead Aft of puck		1	260.2	0.2602	167.6	0.718	-10.20	-0.09365	0.000	0
Tape for trim lead on bottom panel		3	2.4	0.0024	157.5	0.617	-10.20	-0.09365	0.000	0

Appendix D: 25% wing

SEAGLIDER - TRIM AND BALANCE

SG 194

Date

24-Oct-13

Vehicle CG

x_CG

95.752

cm

z_CG

-0.835

cm

y_CG

0

cm

Description	Part #	Qty	Total Weight	Total Weight	Long. CG (x)	Delta_x	VCG&B (z)	Delta_z	Lat. CG (y)	Delta_y
			grams	kg	cm	m	cm	m	cm	m
Tubes & glue mass on top hatch (4) - non-compressee	NIB	1	71.5	0.0715	154.9	0.591	-13.10	-0.12265	0.000	0
Compressee only on top hatch (4)		1	877.7	0.8777	154.9	0.591	-13.10	-0.12265	0.000	0
Tubes & glue mass on bottom hatch (2)		1	36	0.036	154.9	0.591	-12.70	-0.11865	0.000	0
Compressee only on bottom hatch (2)		1	403.1	0.4031	154.9	0.591	-12.70	-0.11865	0.000	0
Tubes & glue mass under port wing (4)		1	71.4	0.0714	132.1	0.363	0.00	0.00835	0.000	0
Compressee only under port wing (4)		1	876.6	0.8766	132.1	0.363	0.00	0.00835	0.000	0
Tubes & glue mass under starboard wing (4)		1	71.2	0.0712	132.1	0.363	0.00	0.00835	0.000	0
Compressee only under starboard wing (4)		1	896.6	0.8966	132.1	0.363	0.00	0.00835	0.000	0
Tubes & glue mass aft on port side (2)		1	35.6	0.0356	161.3	0.655	0.00	0.00835	0.000	0
Compressee only aft on port side (2)		1	385.8	0.3858	161.3	0.655	0.00	0.00835	0.000	0
Tubes aft & glue mass aft on starboard side (2)		1	35.6	0.0356	161.3	0.655	0.00	0.00835	0.000	0
Compressee only aft on starboard side (2)		1	427.1	0.4271	161.3	0.655	0.00	0.00835	0.000	0
Antenna Assembly GPSI (All parts), including extra comms cable	52336 / 52341 / 52454 / 52476 / 55361	1	628.7	0.6287	211.84	1.161	0.00	0.00835	0.000	0
	52220 / 52485 / 52486 / 55239 / 55321	1	293.0	0.293	142.9	0.471	15.90	0.16735	0.000	0
O2 Sensor (Aanderas optode)		1	293.0	0.293	142.9	0.471	15.90	0.16735	0.000	0
O2 Sensor Assy. Hardware (mount base, SS hose clamp, (2x) 8-32 x 5/8 FHMS, (2x) 8-32 nylok)	52220 / 52485 / 52486 / 55239 / 55321	1	34.7	0.0347	142.9	0.471	15.90	0.16735	0.000	0
Wet Labs Sensor		1	234.7	0.2347	154.9	0.591	-10.20	-0.09365	0.000	0
Wet Labs sensor Assy. Hardware ((2x) snesor clamp, 2(x) 8-32 x 5/8 FHMS - sensor clamp to fairing)	55381 / 52312 / 55245	1	3.2	0.0032	154.9	0.591	-10.20	-0.09365	0.000	0
Position D (SBE-O2): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-2.90	-0.02065	0.000	0
Position E (WL-2): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.10	-0.04265	0.000	0
Position F (WL-3 / PAR): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.80	-0.04965	0.000	0
Position G (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.10	-0.04265	0.000	0
Position H (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-2.90	-0.02065	0.000	0
Position I (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	2.90	0.03735	0.000	0
O-Ring E70-270, Endcap to Batt Hull	52411	1	8.0	0.008	123.3	0.275	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS w/ LW + (5x) 6-32 x 5/8 with zincs, endcap to battery hull	55183 / 55217 / 55214 / 49884	1	13.2	0.0132	123.5	0.277	0.00	0.00835	0.000	0
O-Ring, E70-270, Bulkhead to batt. hull	52411	1	8.0	0.008	72.9	-0.229	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS w/ LW + (5x) 6-32 x 5/8 with zincs, bulkhead to battery hull	55183 / 55217 / 55214 / 49884	1	15.3	0.0153	65.4	-0.304	0.00	0.00835	0.000	0
(4x) 6-32 x 1/2 SHCS + #6 LW, rails to bulkhead	55183 / 55217	1	5.3	0.0053	59.5	-0.363	0.00	0.00835	0.000	0
O-Ring, E70-263, electric hull to bulkhead	52410	1	6.4	0.0064	66.4	-0.294	0.00	0.00835	0.000	0
(8x) 6-32 x 1/2 SHCS + #6 LW, elec hull to bulkhead	55183 / 55217	1	10.5	0.0105	65.8	-0.300	0.00	0.00835	0.000	0
(8x) Ogive fairing adapter clips		1	43.7	0.0437	65.8	-0.300	0.00	0.00835	0.000	0

Appendix D: 25% wing

SEAGLIDER - TRIM AND BALANCE

SG 194

Date

24-Oct-13

Vehicle CG

x_CG

95.752

cm

z_CG

-0.835

cm

y_CG

0

cm

Description	Part #	Qty	Total Weight	Total Weight	Long. CG (x)	Delta_x	VCG&B (z)	Delta_z	Lat. CG (y)	Delta_y
			grams	kg	cm	m	cm	m	cm	m
(8x) 1/4-28 x 3/8 PHMS, Ogive fairing adapter clip to aft endcap		1	37.1	0.0371	65.8	-0.300	0.00	0.00835	0.000	0
Endcap, Aft	49815	1	2027.6	2.0276	127.0	0.312	0.00	0.00835	0.000	0
(2x) Roll pin, 5/32 OD x 3/8, Pump anti rotation	55248	1	1.0	0.001	123.0	0.272	0.00	0.00835	0.000	0
Connector, Run Tee, 1/8 NPT x 1/4 tube, Push to connect		1	8.4	0.0084	125.2	0.294	0.00	0.00835	0.000	0
Connector, Male, 7/16 straight x 1/4 tube, SS	55506	1	27.6	0.0276	125.2	0.294	1.30	0.02135	0.000	0
Assy. Solenoid Valve, Skinner	52254	1	600.7	0.6007	124.5	0.287	1.50	0.02335	0.000	0
Adaptor, Male, 1/8 NPT x 1/4 M. SS	52430	1	10.5	0.0105	128.1	0.323	1.30	0.02135	0.000	0
Connector, Male, 1/4 tube x 1/8 NPT, Push to connect		1	6.3	0.0063	123.9	0.281	-1.30	-0.00465	0.000	0
(3x) Tubing, 1/4 DIA x 10"		1	12.1	0.0121	119.4	0.236	0.00	0.00835	0.000	0
(2x) Tubing, 1/4 Dia x 4.375"		1	3.5	0.0035	119.4	0.236	0.00	0.00835	0.000	0
Tubing, 1/4 Dia x 1.25"		1	0.5	0.0005	119.4	0.236	0.00	0.00835	0.000	0
Tubing, 1/4 DIA x 8"		1	3.2	0.0032	119.4	0.236	0.00	0.00835	0.000	0
Check Valve, Kepner 204 A-1	52448	1	23.0	0.023	124.7	0.289	-1.30	-0.00465	0.000	0
Branch Tee, Male, 1/4 tube x 1/8 NPT, Push to connect		1	8.4	0.0084	125.2	0.294	0.00	0.00835	0.000	0
Seal Washer	52350	1	0.9	0.0009	129.3	0.335	0.00	0.00835	0.000	0
LeDuc Output Screw	52349	1	7.8	0.0078	129.6	0.338	0.00	0.00835	0.000	0
Hydraulic Pump, LeDuc	52348	1	317.2	0.3172	129.6	0.338	0.00	0.00835	0.000	0
Hydraulic endcap	49803	1	111.1	0.1111	125.9	0.301	0.00	0.00835	0.000	0
Ball Bearing, 8 mm ID x 16 mm OD	52459	1	3.1	0.0031	121.2	0.254	0.00	0.00835	0.000	0
O-Ring, N70-032, hydraulic endcap	52407	1	0.5	0.0005	121.1	0.253	0.00	0.00835	0.000	0
Connector, Male, 1/16 NPT x 1/8 tube, Push to connect		1	5.9	0.0059	122.5	0.267	2.90	0.03735	0.000	0
Tubing, 1/8 Dia. x 10.5"	52418	1	0.9	0.0009	121.3	0.255	0.00	0.00835	0.000	0
Magnetic Seal	52353	1	13.9	0.0139	114.7	0.189	0.00	0.00835	0.000	0
(6x) 6-32 x 1/2 SHCS + #6 LW, hydraulic encap to encap	55183 / 55217	1	7.8	0.0078	121.1	0.253	0.00	0.00835	0.000	0
Lee Plug Set	52445 / 52446	1	0.4	0.0004	122.2	0.264	-1.30	-0.00465	0.000	0
Check Valve (altered item Kepner 1106 A-1-1) + O-Ring (N70 3-906)	49852 / 52408	1	48.1	0.0481	130.8	0.350	0.00	0.00835	0.000	0
Bladder, External	52441	1	837.8	0.8378	133.3	0.375	0.00	0.00835	0.000	0
Nut, Bladder	49802	1	107.3	0.1073	138.7	0.429	0.00	0.00835	0.000	0
Position A (Comms): IE55 w o-ring	52160	1	11.7	0.0117	131.1	0.353	5.10	0.05935	0.000	0
Position B (Opt O2): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	2.90	0.03735	0.000	0
Position C (WL-1): IE55 w o-ring	52160	1	11.5	0.0115	130.8	0.350	0.00	0.00835	0.000	0
Position D (SBE O2): IE55 w o-ring	52160	1	11.8	0.0118	130.8	0.350	-2.90	-0.02065	0.000	0
Position E (WL-2): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.10	-0.04265	0.000	0
Position F (WL-3): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.80	-0.04965	0.000	0
Position G (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.10	-0.04265	0.000	0
Position H (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-2.90	-0.02065	0.000	0
Position I (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	2.90	0.03735	0.000	0

Appendix D: 25% wing										
SEAGLIDER - TRIM AND BALANCE		SG 194	Date	24-Oct-13						
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Position J (?): IE55 w o-ring	52160	1	11.8	0.0118	130.8	0.350	5.10	0.05935	0.000	0
Iridium/GPS Ant., conn/o-ring/ CA W5, J2	52241	1	66.1	0.0661	130.8	0.350	0.00	0.00835	0.000	0
Vaccum/pressure relief valve (deep sea power)	52449	1	11.5	0.0115	130.8	0.350	5.80	0.06635	0.000	0
Pump Plate, Inner	49804	1	37.8	0.0378	120.1	0.243	1.90	0.02735	0.000	0
Assy. Main pump motor w/ CA W8, W18	52253	1	130.7	0.1307	123.2	0.274	4.60	0.05435	0.000	0
(3x) M2 x 6 FHMS (motor to interface plate)	55203	1	0.4	0.0004	120.1	0.243	4.60	0.05435	0.000	0
Bore Reducer	52351	1	0.7	0.0007	118.8	0.230	0.00	0.00835	0.000	0
Pulley, LeDuc pump motor	52352	1	5.6	0.0056	118.8	0.230	4.70	0.05535	0.000	0
(6x) 4-40 x 3/8 FHMS, hyd end cap to inner plate	55229	1	1.9	0.0019	120.1	0.243	0.00	0.00835	0.000	0
Pulley, LeDuc pump	49806	1	91.7	0.0917	118.8	0.230	0.00	0.00835	0.000	0
Timing Belt	52357	1	3.4	0.0034	118.8	0.230	1.30	0.02135	0.000	0
Pump Plate, Outer w/ (4x) standoffs	49805	1	20.2	0.0202	118.1	0.223	1.90	0.02735	0.000	0
(4x) 4-40 x 3/16 FHCS		1	1.2	0.0012	118.4	0.226	0.00	0.00835	0.000	0
Check Valve, Kepner 404A-1		1	25.3	0.0253	125.2	0.294	0.00	0.00835	0.000	0
Connector, 45 elbow, 1/8 NPT x 1/4 tube, Push to connect		1	7.2	0.0072	125.2	0.294	0.00	0.00835	0.000	0
Connector, 90 elbow, 1/8 NPT x 1/4 tube, Push to connect		1	7.5	0.0075	125.2	0.294	0.00	0.00835	0.000	0
Connector, Union Tee, 1/4 tube, Push to connect		1	2.9	0.0029	125.2	0.294	0.00	0.00835	0.000	0
Pump Bracket		1	21.5	0.0215	124.2	0.284	0.00	0.00835	0.000	0
Motor, Chrg Pump w/ W34		1	59.9	0.0599	124.2	0.284	0.00	0.00835	0.000	0
(2x) M2 x 6 FHMS, motor to bracket		1	0.2	0.0002	124.2	0.284	2.50	0.03335	0.000	0
Magnet Rotor Housing		1	10.5	0.0105	124.2	0.284	0.00	0.00835	0.000	0
(4x) 6-32 x 1/4 PHMS, bracket to housing		1	1.8	0.0018	124.2	0.284	-1.00	-0.00165	0.000	0
Rotary Magnet		1	18.0	0.018	124.2	0.284	-1.00	-0.00165	0.000	0
Pumphead, Micropump + mounting plate + (3x) 4-40 x 3/8 FHMS		1	224.7	0.2247	124.2	0.284	-2.00	-0.01165	0.000	0
(4x) 4-40 x 3/16 SHCS + #4 LW, housing to flange		1	2.0	0.002	124.2	0.284	-1.30	-0.00465	0.000	0
(2x) Elbow, 1/8 not x 1/4 tube, push to connect		1	15.0	0.015	124.2	0.284	-2.30	-0.01465	0.000	0
Piston, Diaphragm	49811	1	135.3	0.1353	114.3	0.185	0.00	0.00835	0.000	0
Diaphragm, Internal (Bellofram)	52359	1	551.9	0.5519	115.7	0.199	0.00	0.00835	0.000	0
Sikaflex Adhesive Compound	55459	1	15.6	0.0156	114.3	0.185	0.00	0.00835	0.000	0
Cylinder Head, Diaphragm with extra port		1	336.4	0.3364	111.5	0.157	0.00	0.00835	0.000	0
Branch Tee, 1/4 tube x 1/8 NPT, push to connect; 1/4 tube plug		1	9.2	0.0092	109.8	0.140	-6.10	-0.05265	0.000	0
Elbow, 1/8 NPT M x 1/8 NPT M, brass	52426	1	26.9	0.0269	109.8	0.140	6.10	0.06935	0.000	0
Kepner Check Valve 404A-1-200		1	26.6	0.0266	109.8	0.140	2.00	0.02835	0.000	0
90 Elbow, 1/8 npt x 1/8 tube, push to connect		1	7.4	0.0074	109.8	0.140	-0.30	0.00535	0.000	0
Filter, Baston (25 micron) / with zip tie		1	20.9	0.0209	109.8	0.140	2.00	0.02835	0.000	0
Union, 1/4 tube, push to connect		1	1.8	0.0018	109.8	0.140	2.00	0.02835	0.000	0
90 Elbow, 1/8 npt x 1/4 tube, push to connect		1	7.5	0.0075	109.8	0.140	2.00	0.02835	0.000	0

Appendix D: 25% wing

SEAGLIDER - TRIM AND BALANCE

SG 194

Date

24-Oct-13

Vehicle CG

x_CG

95.752

cm

z_CG

-0.835

cm

y_CG

0

cm

Description	Part #	Qty	Total Weight	Total Weight	Long. CG (x)	Delta_x	VCG&B (z)	Delta_z	Lat. CG (y)	Delta_y
			grams	kg	cm	m	cm	m	cm	m
Bleed screw, 10-32 x .125 PHMS with seal	55477	1	1.8	0.0018	109.8	0.140	7.60	0.08435	0.000	0
Cylinder, Diaphragm	49812	1	421.5	0.4215	115.1	0.193	0.00	0.00835	0.000	0
(8x) 5-40 x 3/8 SHCS + #5 LW, head to cylinder	55251	1	7.5	0.0075	110.6	0.148	0.00	0.00835	0.000	0
Standoff (starboard)	55299	1	12.2	0.0122	121.3	0.255	6.60	0.07435	0.000	0
Standoff (port)	55299	1	12.5	0.0125	121.3	0.255	-6.60	-0.05765	0.000	0
(4x) 6-32 x 3/8 SHCS, standoff to cylinder	55183	1	4.5	0.0045	120.7	0.249	0.00	0.00835	0.000	0
(2x) Saddles, mounting, Aft PCB + (4x) 4-40 x 1/4 PHMS	52373 / 55257	1	5.3	0.0053	115.6	0.198	8.10	0.08935	0.000	0
(2x) Saddles, mounting, C PCB + (4x) 4-40 x 1/4 PHMS	49888 / 55257	1	6.7	0.0067	115.6	0.198	3.80	0.04635	0.000	0
(2x) Saddles, mounting, T PCB + (4x) 4-40 x 1/4 PHMS	49889 / 55257	1	6.7	0.0067	115.6	0.198	-3.80	-0.02965	0.000	0
(2x) Saddles, mounting, SBE O2 PCB + (4x) 4-40 x 1/4 SHCS	49890 / 55257	1	4.5	0.0045	115.6	0.198	-8.10	-0.07265	0.000	0
Linear Poteniometer A w/ cable assy (W21), Linear pot clamp, 3-48 x 1/2 SHCS, Spring, Linear pot rod end	52255 / 49814 / 55220	1	40.9	0.0409	121.3	0.255	6.40	0.07235	0.000	0
Linear Potentiometer B w/ cable assy (W20), Linear pot clamp, 3-48 x 1/2 SHCS, Spring, Linear pot rod end	52256 / 49814 / 55220	1	38.5	0.0385	121.3	0.255	6.40	0.07235	0.000	0
(2x) 5-40 x 1/2 SHCS + #5 LW, Clamp to cylinder	55250 / 55212 / 55225	1	1.9	0.0019	121.0	0.252	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS + #6 LW, boost pump bracket to endcap	55183 / 55217	1	2.7	0.0027	122.6	0.268	0.00	0.00835	0.000	0
(4x) 6-32 x 1/2 SHCS + #6 LW, cylinder standoff to endcap	55184 / 55217	1	5.4	0.0054	121.7	0.259	0.00	0.00835	0.000	0
Aft Terminal PCB	55302	1	25.0	0.025	115.6	0.198	8.60	0.09435	0.000	0
SBE Conductivity PCB (included with sensors)	55385	1	32.6	0.0326	115.6	0.198	4.30	0.05135	0.000	0
Conductivity, PCB jumper, CA W26	52261	1	3.1	0.0031	112.8	0.170	3.80	0.04635	0.000	0
Shield, Radiation, C PCB + (4x) 2-56 x 1/4 BHCS	49887 / 55221	1	20.1	0.0201	115.6	0.198	4.80	0.05635	0.000	0
SBE Temperature PCB (included with sensors)	55388	1	22.0	0.022	115.6	0.198	-4.30	-0.03465	0.000	0
Temperature, PCB jumper, CA W25	52260	1	5.2	0.0052	112.8	0.170	1.30	0.02135	0.000	0
Shield, Radiation, T PCB + (4x) 2-56 x 1/4 BHCS	52371 / 55221	1	16.5	0.0165	115.6	0.198	-4.80	-0.03965	0.000	0
Oxygen, PCB jumper, CA W33 (only used with SBE 43f)	52265	1	6.1	0.0061	112.8	0.170	0.00	0.00835	0.000	0
Ribbon Cable (W13) (gray)	52248	1	29.6	0.0296	98.1	0.023	8.90	0.09735	0.000	0
Ribbon Cable (W14) (rainbow)	52249	1	37.1	0.0371	98.1	0.023	8.90	0.09735	0.000	0
CT Sensor Sail Assembly (stanchion, guard, nut, nacelle, (6x) 6-32 x 5/16 FHMS, o-ring (E70-016), o-ring (E70-012), SBE components)	52368	1	293.8	0.2938	121.6	0.258	17.90	0.18735	0.000	0
Mass Shifter Stability Rod Assy w/ Hard Stop ("the stinger")		1	87.6	0.0876	111.5	0.157	0.00	0.00835	0.000	0
Electronic rail, port	49846	1	60.9	0.0609	59.7	-0.361	0.00	0.00835	0.000	0
Electronics rail, starboard	52379	1	60.9	0.0609	59.7	-0.361	0.00	0.00835	0.000	0
Seaglider Mainboard (Mother board, TT8, CF8, flash card)	522990-B.4	1	263.5	0.2635	54.6	-0.412	0.70	0.01535	0.000	0
Insulation Sheet-OEM/Main	52370	1	4.0	0.004	55.2	-0.406	0.44	0.01275	0.000	0
(6x) 6-32 x 1/4 PHMS, main board to rails	55237	1	4.5	0.0045	54.6	-0.412	0.50	0.01335	0.000	0

Appendix D: 25% wing

SEAGLIDER - TRIM AND BALANCE	SG 194	Date	24-Oct-13							
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Assy, OEM Navigation Board (transponder PCB, transformer/inductor, capacitors, GPS receiver, W6 GPS serial interface, GPS antenna cable, DC block, Cable Assembly W7, all soldered on wiring) + RAFOS clock and receiver board & cable to mainboard		1	559.1	0.5591	54.6	-0.412	-0.43	0.00405	0.000	0
(6x) 6-32 x 1/4 PHMS, OEM board to rails	55237	1	4.5	0.0045	54.6	-0.412	-0.43	0.00405	0.000	0
Assy, Iridium Moedm and bracket (includes satellite reciever - iridium LBT, cable assy W3, all clamps for wires, RF switch, DC block, and mounting hardware)	52324	1	271.6	0.2716	54.9	-0.409	1.90	0.02735	0.000	0
4x) 6-32 x 1/4 PHMS + #6 LW, modem cage to rails	55209 / 55217	1	3.4	0.0034	52.8	-0.430	0.29	0.01125	0.000	0
Battery Pack, LV w/ CA W1	52237	1	2664.4	2.6644	54.9	-0.409	-1.81	-0.00975	0.000	0
Battery Pack Cage, LV	49847	1	71.6	0.0716	54.9	-0.409	-1.64	-0.00805	0.000	0
(6x) 6-32 x 1/4 PHMS + #6 LW, batt. cage to rails	55209 / 55217	1	5.4	0.0054	55.3	-0.405	-0.25	0.00585	0.000	0
Hull, Fwd Battery	49817	1	3646.0	3.646	110.6	0.148	0.00	0.00835	0.000	0
Hull, Aft Battery	49817	1	3638.6	3.6386	85.5	-0.103	0.00	0.00835	0.000	0
Joint Ring, battery hulls	49818	1	674.7	0.6747	98.9	0.031	0.00	0.00835	0.000	0
(8x) 6-32 x 3/4 SHCS + #6 LW, joint ring to fwd/aft batt. Hull	55214 / 55217	1	13.3	0.0133	98.1	0.023	0.00	0.00835	0.000	0
(2x) O-Ring, E70-271	52412	1	17.1	0.0171	98.1	0.023	0.00	0.00835	0.000	0
(12x) Clips, Rib	49883	1	25.4	0.0254	98.1	0.023	3.95	0.04785	0.000	0
(12x) Clips, Wiring	52405	1	18.7	0.0187	98.1	0.023	3.95	0.04785	0.000	0
Oil Absorption pad, 1.9" x 10.125" + (2x) rib clips	55527 / 49883	1	11.2	0.0112	98.1	0.023	3.95	0.04785	0.000	0
Endcap, Forward	49830	1	1897.1	1.8971	39.5	-0.563	0.00	0.00835	0.000	0
Acoustic transducer (ITC-3013), w/ cable W36	52267 / 52223	1	1438.7	1.4387	22.9	-0.729	0.00	0.00835	0.000	0
O-Ring, E70-238, transducer to Endcap	52413	1	3.2	0.0032	26.7	-0.691	0.00	0.00835	0.000	0
(6x) Alum 1/4-20 x 3/4 SHCS + 1/4 LW, xducer to end cap	55236 / 55252	1	15.2	0.0152	27.2	-0.686	0.00	0.00835	0.000	0
Hull, Electronics	49829	1	1696.0	1.696	58.0	-0.378	0.00	0.00835	0.000	0
O-Ring, E70-263, elec hull to fwd endcap	52410	1	7.1	0.0071	50.2	-0.456	0.00	0.00835	0.000	0
(8x) 6-32 x 1/2 SHCS + #6 LW, electric hull to fwd encap	55183 / 55217	1	10.3	0.0103	50.5	-0.453	0.00	0.00835	0.000	0
Compass bracket	49848	1	24.4	0.0244	30.5	-0.653	-0.35	0.00485	0.000	0
Compass w/compass carrier PCB	55305 / 52466	1	46.8	0.0468	30.5	-0.653	0.00	0.00835	0.000	0
(4x) 4-40 x 3/8 SHCS + #4 x 1/8L stand off	55256 / 55282	1	2.6	0.0026	30.5	-0.653	-0.31	0.00525	0.000	0
(2x) 4-40 x 3/8 SHCS + #4 LW, comp mt to fwd encap	55228 / 55233	1	1.3	0.0013	27.4	-0.684	-0.81	0.00025	0.000	0
Cable Assembly W9 (compass cable)	52244	1	21.5	0.0215	38.9	-0.569	0.38	0.01215	0.000	0
Total weight (SUM of components' weight col D)			44255	44.255	Excluding movable battery mass					
Total Scale Weight			58852							

Appendix E: 0% rudder

SEAGLIDER - TRIM AND BALANCE			SG 194	Date	24-Oct-13						
	Vehicle CG	x_CG	95.752		cm						
		z_CG	-0.835		cm						
		y_CG	0		cm						
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m	
(8x) 10-32 x 1/2 FHMS, joint ring to aft fairing	55279	1	14.3	0.0143	116.8	0.210	0.00	0.00835	0.000	0	
(8x) 10-32 x 1/2 FHMS, Ogive Aft fairing to endcap	55279	1	14.3	0.0143	125.7	0.299	0.00	0.00835	0.000	0	
Forward Fairing - with foam	49897	1	4237.1	4.2371	71.7	-0.241	0.00	0.00835	0.000	0	
Joint Ring, Fwd to Aft Fairing (bonded to fwd fairing before paint; Long ring = 875g, short ring 688g)	49834	1	688	0.688	116.5	0.207	0.00	0.00835	0.000	0	
Nose weight (bonded to fwd fairing after paint)	49844	1	1270.9	1.2709	9.4	-0.864	0.00	0.00835	0.000	0	
Nose weight plates	55448	1	495.3	0.4953	9.4	-0.864	0.00	0.00835	0.000	0	
(2x) 1/4-20 x 1.5" SHCS + 1/4 LW brass, nose plate to nose base (2 -3 plates)	55185/55253	1	23.4	0.0234	9.4	-0.864	0.00	0.00835	0.000	0	
Trim Lead .600" aft of pupa joint ring - bottom (5" strip)		1	695.1	0.6951	106.9	0.111	-13.10	-0.12265	0.000	0	
EPDM base (5"x6") + (2x) 2"x 6" tape - bottom		1	24.2	0.0242	106.9	0.111	-12.70	-0.11865	0.000	0	
Trim Lead .600" aft of pupa joint ring - port side (5" strip)		1	759.8	0.7598	106.9	0.111	0.00	0.00835	0.000	0	
EPDM base (5"x6") + (2x) 2"x 6" tape - port		1	24.0	0.024	106.9	0.111	0.00	0.00835	0.000	0	
Trim Lead 600" aft of pupa joint ring - starboard side (5" strip)		1	865.0	0.865	106.9	0.111	0.00	0.00835	0.000	0	
EPDM base (5x6") + (2x) 2"x 6" tape - starboard		1	24.0	0.024	106.9	0.111	0.00	0.00835	0.000	0	
Trim Lead tape (~1g per 5" lead strip)		15	12.0	0.012	106.9	0.111	13.00	0.13835	0.000	0	
(2) Straps - cut to length (start at 34g, subtract trimmed)		1	33.2	0.0332	106.9	0.111	0.00	0.00835	0.000	0	
(2) wraps 2x around black elec tape for security		1	12.0	0.012	106.9	0.111	0.00	0.00835	0.000	0	
Ogive Aft Fairing - with cutouts for panels	56972	1	2535	2.535	143.2	0.474	-0.10	0.00735	0.000	0	
Ogive Wing, Starboard (1m)	49838	1	617.8	0.6178	129.779	0.340	0.00	0.00835	29.905	0.29905	
Ogive Wing, Port (1 m)	49838	1	611.3	0.6113	129.779	0.340	0.00	0.00835	-29.905	-0.29905	
(8x) 10-32 x 1/2 FHMS, wings to fairing	5xxxx	1	14.3	0.0143	129.6	0.338	0.00	0.00835	0.000	0	
Top Rudder			0	0.0627	0	0.750	0.00	0.10079	0	0	
Bottom Rudder		1	62.7	0.0627	170.784	0.750	-10.91	-0.10079	0.000	0	
(2x) 1/4-20 x 1 1/2" FHMS, rudder to rudder shoe	5xxxx	1	15.7	0.0157	181.8	0.860	0.00	0.00835	0.000	0	
Ogive Top panel (with sensor holes)	57651	1	298.1	0.2981	142.8	0.470	11.80	0.12635	0.000	0	
Bottom panel with sensor holes	57651	1	290.4	0.2904	142.8	0.470	-11.80	-0.10965	0.000	0	
(20x) Ogive Panel screws - 6-32 x 3/8 Truss head MS	56912	1	19.2	0.0192	142.8	0.470	0.00	0.00835	0.000	0	
Top panel tie-downs for compresssee (10-32x5/8 FHMS, nut, anchor)	NIB	4	14.8	0.0148	132.1	0.363	13.30	0.14135	0.000	0	
Bottom panel tie-downs for compresssee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	143	0.472	13.30	0.14135	0.000	0	
Aft-end strb Tie-downs for compresssee (10-32x5/8 FHMS, nut, anchor)		2	7.4	0.0074	161.3	0.655	0.00	0.00835	0.000	0	
Aft-end port Tie-downs for compresssee (10-32x5/8 FHMS, nut, anchor)		2	7.4	0.0074	161.3	0.655	0.00	0.00835	0.000	0	
Under strb-wing Tie-downs for compresssee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	132.1	0.363	0.00	0.00835	0.000	0	
Under port-wing Tie-downs for compresssee (10-32x5/8 FHMS, nut, anchor)		4	14.8	0.0148	132.1	0.363	0.00	0.00835	0.000	0	
Trim lead to Port & Starbord of puck		1	526	0.526	154.9	0.591	-10.20	-0.09365	0.000	0	
Trim lead Aft of puck		1	260.2	0.2602	167.6	0.718	-10.20	-0.09365	0.000	0	
Tape for trim lead on bottom panel		3	2.4	0.0024	157.5	0.617	-10.20	-0.09365	0.000	0	

Appendix E: 0% rudder										
SEAGLIDER - TRIM AND BALANCE			SG 194	Date	24-Oct-13					
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Tubes & glue mass on top hatch (4) - non-compressee	NIB	1	71.5	0.0715	154.9	0.591	-13.10	-0.12265	0.000	0
Compressee only on top hatch (4)		1	877.7	0.8777	154.9	0.591	-13.10	-0.12265	0.000	0
Tubes & glue mass on bottom hatch (2)		1	36	0.036	154.9	0.591	-12.70	-0.11865	0.000	0
Compressee only on bottom hatch (2)		1	403.1	0.4031	154.9	0.591	-12.70	-0.11865	0.000	0
Tubes & glue mass under port wing (4)		1	71.4	0.0714	132.1	0.363	0.00	0.00835	0.000	0
Compressee only under port wing (4)		1	876.6	0.8766	132.1	0.363	0.00	0.00835	0.000	0
Tubes & glue mass under starboard wing (4)		1	71.2	0.0712	132.1	0.363	0.00	0.00835	0.000	0
Compressee only under starboard wing (4)		1	896.6	0.8966	132.1	0.363	0.00	0.00835	0.000	0
Tubes & glue mass aft on port side (2)		1	35.6	0.0356	161.3	0.655	0.00	0.00835	0.000	0
Compressee only aft on port side (2)		1	385.8	0.3858	161.3	0.655	0.00	0.00835	0.000	0
Tubes aft & glue mass aft on starboard side (2)		1	35.6	0.0356	161.3	0.655	0.00	0.00835	0.000	0
Compressee only aft on starboard side (2)		1	427.1	0.4271	161.3	0.655	0.00	0.00835	0.000	0
Antenna Assembly GPSI (All parts), including extra comms cable	52336 / 52341 / 52454 / 52476 / 55361	1	628.7	0.6287	211.84	1.161	0.00	0.00835	0.000	0
	52220 / 52485 / 52486 / 55239 / 55321	1	293.0	0.293	142.9	0.471	15.90	0.16735	0.000	0
O2 Sensor (Aanderas optode)										
O2 Sensor Assy. Hardware (mount base, SS hose clamp, (2x) 8-32 x 5/8 FHMS, (2x) 8-32 nylok)	52220 / 52485 / 52486 / 55239 / 55321	1	34.7	0.0347	142.9	0.471	15.90	0.16735	0.000	0
Wet Labs Sensor		1	234.7	0.2347	154.9	0.591	-10.20	-0.09365	0.000	0
Wet Labs sensor Assy. Hardware ((2x) snesor clamp, 2(x) 8-32 x 5/8 FHMS - sensor clamp to fairing)	55381 / 52312 / 55245	1	3.2	0.0032	154.9	0.591	-10.20	-0.09365	0.000	0
Position D (SBE-O2): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-2.90	-0.02065	0.000	0
Position E (WL-2): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.10	-0.04265	0.000	0
Position F (WL-3 / PAR): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.80	-0.04965	0.000	0
Position G (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-5.10	-0.04265	0.000	0
Position H (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	-2.90	-0.02065	0.000	0
Position I (?): IE55 dummy plug	55309	1	7.3	0.0073	131.4	0.356	2.90	0.03735	0.000	0
O-Ring E70-270, Endcap to Batt Hull	52411	1	8.0	0.008	123.3	0.275	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS w/ LW + (5x) 6-32 x 5/8 with zincs, endcap to battery hull	55183 / 55217 / 55214 / 49884	1	13.2	0.0132	123.5	0.277	0.00	0.00835	0.000	0
O-Ring, E70-270, Bulkhead to batt. hull	52411	1	8.0	0.008	72.9	-0.229	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS w/ LW + (5x) 6-32 x 5/8 with zincs, bulkhead to battery hull	55183 / 55217 / 55214 / 49884	1	15.3	0.0153	65.4	-0.304	0.00	0.00835	0.000	0
(4x) 6-32 x 1/2 SHCS + #6 LW, rails to bulkhead	55183 / 55217	1	5.3	0.0053	59.5	-0.363	0.00	0.00835	0.000	0
O-Ring, E70-263, electric hull to bulkhead	52410	1	6.4	0.0064	66.4	-0.294	0.00	0.00835	0.000	0
(8x) 6-32 x 1/2 SHCS + #6 LW, elec hull to bulkhead	55183 / 55217	1	10.5	0.0105	65.8	-0.300	0.00	0.00835	0.000	0
(8x) Ogive fairing adapter clips		1	43.7	0.0437	65.8	-0.300	0.00	0.00835	0.000	0

Appendix E: 0% rudder										
SEAGLIDER - TRIM AND BALANCE		SG 194	Date	24-Oct-13						
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
(8x) 1/4-28 x 3/8 PHMS, Ogive fairing adapter clip to aft endcap		1	37.1	0.0371	65.8	-0.300	0.00	0.00835	0.000	0
Endcap, Aft	49815	1	2027.6	2.0276	127.0	0.312	0.00	0.00835	0.000	0
(2x) Roll pin, 5/32 OD x 3/8, Pump anti rotation	55248	1	1.0	0.001	123.0	0.272	0.00	0.00835	0.000	0
Connector, Run Tee, 1/8 NPT x 1/4 tube, Push to connect		1	8.4	0.0084	125.2	0.294	0.00	0.00835	0.000	0
Connector, Male, 7/16 straight x 1/4 tube, SS	55506	1	27.6	0.0276	125.2	0.294	1.30	0.02135	0.000	0
Assy, Solenoid Valve, Skinner	52254	1	600.7	0.6007	124.5	0.287	1.50	0.02335	0.000	0
Adaptor, Male, 1/8 NPT x 1/4 M. SS	52430	1	10.5	0.0105	128.1	0.323	1.30	0.02135	0.000	0
Connector, Male, 1/4 tube x 1/8 NPT, Push to connect		1	6.3	0.0063	123.9	0.281	-1.30	-0.00465	0.000	0
(3x) Tubing, 1/4 DIA x 10"		1	12.1	0.0121	119.4	0.236	0.00	0.00835	0.000	0
(2x) Tubing, 1/4 Dia x 4.375"		1	3.5	0.0035	119.4	0.236	0.00	0.00835	0.000	0
Tubing, 1/4 Dia x 1.25"		1	0.5	0.0005	119.4	0.236	0.00	0.00835	0.000	0
Tubing, 1/4 DIA x 8"		1	3.2	0.0032	119.4	0.236	0.00	0.00835	0.000	0
Check Valve, Kepner 204 A-1	52448	1	23.0	0.023	124.7	0.289	-1.30	-0.00465	0.000	0
Branch Tee, Male, 1/4 tube x 1/8 NPT, Push to connect		1	8.4	0.0084	125.2	0.294	0.00	0.00835	0.000	0
Seal Washer	52350	1	0.9	0.0009	129.3	0.335	0.00	0.00835	0.000	0
LeDuc Output Screw	52349	1	7.8	0.0078	129.6	0.338	0.00	0.00835	0.000	0
Hydraulic Pump, LeDuc	52348	1	317.2	0.3172	129.6	0.338	0.00	0.00835	0.000	0
Hydraulic endcap	49803	1	111.1	0.1111	125.9	0.301	0.00	0.00835	0.000	0
Ball Bearing, 8 mm ID x 16 mm OD	52459	1	3.1	0.0031	121.2	0.254	0.00	0.00835	0.000	0
O-Ring, N70-032, hydraulic endcap	52407	1	0.5	0.0005	121.1	0.253	0.00	0.00835	0.000	0
Connector, Male, 1/16 NPT x 1/8 tube, Push to connect		1	5.9	0.0059	122.5	0.267	2.90	0.03735	0.000	0
Tubing, 1/8 Dia. x 10.5"	52418	1	0.9	0.0009	121.3	0.255	0.00	0.00835	0.000	0
Magnetic Seal	52353	1	13.9	0.0139	114.7	0.189	0.00	0.00835	0.000	0
(6x) 6-32 x 1/2 SHCS + #6 LW, hydraulic encap to encap	55183 / 55217	1	7.8	0.0078	121.1	0.253	0.00	0.00835	0.000	0
Lee Plug Set	52445 / 52446	1	0.4	0.0004	122.2	0.264	-1.30	-0.00465	0.000	0
Check Valve (altered item Kepner 1106 A-1-1) + O-Ring (N70 3-906)	49852 / 52408	1	48.1	0.0481	130.8	0.350	0.00	0.00835	0.000	0
Bladder, External	52441	1	837.8	0.8378	133.3	0.375	0.00	0.00835	0.000	0
Nut, Bladder	49802	1	107.3	0.1073	138.7	0.429	0.00	0.00835	0.000	0
Position A (Comms): IE55 w o-ring	52160	1	11.7	0.0117	131.1	0.353	5.10	0.05935	0.000	0
Position B (Opt O2): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	2.90	0.03735	0.000	0
Position C (WL-1): IE55 w o-ring	52160	1	11.5	0.0115	130.8	0.350	0.00	0.00835	0.000	0
Position D (SBE O2): IE55 w o-ring	52160	1	11.8	0.0118	130.8	0.350	-2.90	-0.02065	0.000	0
Position E (WL-2): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.10	-0.04265	0.000	0
Position F (WL-3): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.80	-0.04965	0.000	0
Position G (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-5.10	-0.04265	0.000	0
Position H (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	-2.90	-0.02065	0.000	0
Position I (?): IE55 w o-ring	52160	1	10.6	0.0106	130.8	0.350	2.90	0.03735	0.000	0

Appendix E: 0% rudder										
SEAGLIDER - TRIM AND BALANCE		SG 194	Date	24-Oct-13						
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Position J (?): IE55 w o-ring	52160	1	11.8	0.0118	130.8	0.350	5.10	0.05935	0.000	0
Iridium/GPS Ant., conn/o-ring/ CA W5, J2	52241	1	66.1	0.0661	130.8	0.350	0.00	0.00835	0.000	0
Vaccum/pressure relief valve (deep sea power)	52449	1	11.5	0.0115	130.8	0.350	5.80	0.06635	0.000	0
Pump Plate, Inner	49804	1	37.8	0.0378	120.1	0.243	1.90	0.02735	0.000	0
Assy. Main pump motor w/ CA W8, W18	52253	1	130.7	0.1307	123.2	0.274	4.60	0.05435	0.000	0
(3x) M2 x 6 FHMS (motor to interface plate)	55203	1	0.4	0.0004	120.1	0.243	4.60	0.05435	0.000	0
Bore Reducer	52351	1	0.7	0.0007	118.8	0.230	0.00	0.00835	0.000	0
Pulley, LeDuc pump motor	52352	1	5.6	0.0056	118.8	0.230	4.70	0.05535	0.000	0
(6x) 4-40 x 3/8 FHMS, hyd end cap to inner plate	55229	1	1.9	0.0019	120.1	0.243	0.00	0.00835	0.000	0
Pulley, LeDuc pump	49806	1	91.7	0.0917	118.8	0.230	0.00	0.00835	0.000	0
Timing Belt	52357	1	3.4	0.0034	118.8	0.230	1.30	0.02135	0.000	0
Pump Plate, Outer w/ (4x) standoffs	49805	1	20.2	0.0202	118.1	0.223	1.90	0.02735	0.000	0
(4x) 4-40 x 3/16 FHCS		1	1.2	0.0012	118.4	0.226	0.00	0.00835	0.000	0
Check Valve, Kepner 404A-1		1	25.3	0.0253	125.2	0.294	0.00	0.00835	0.000	0
Connector, 45 elbow, 1/8 NPT x 1/4 tube, Push to connect		1	7.2	0.0072	125.2	0.294	0.00	0.00835	0.000	0
Connector, 90 elbow, 1/8 NPT x 1/4 tube, Push to connect		1	7.5	0.0075	125.2	0.294	0.00	0.00835	0.000	0
Connector, Union Tee, 1/4 tube, Push to connect		1	2.9	0.0029	125.2	0.294	0.00	0.00835	0.000	0
Pump Bracket		1	21.5	0.0215	124.2	0.284	0.00	0.00835	0.000	0
Motor, Chrg Pump w/ W34		1	59.9	0.0599	124.2	0.284	0.00	0.00835	0.000	0
(2x) M2 x 6 FHMS, motor to bracket		1	0.2	0.0002	124.2	0.284	2.50	0.03335	0.000	0
Magnet Rotor Housing		1	10.5	0.0105	124.2	0.284	0.00	0.00835	0.000	0
(4x) 6-32 x 1/4 PHMS, bracket to housing		1	1.8	0.0018	124.2	0.284	-1.00	-0.00165	0.000	0
Rotary Magnet		1	18.0	0.018	124.2	0.284	-1.00	-0.00165	0.000	0
Pumphead, Micropump + mounting plate + (3x) 4-40 x 3/8 FHMS		1	224.7	0.2247	124.2	0.284	-2.00	-0.01165	0.000	0
(4x) 4-40 x 3/16 SHCS + #4 LW, housing to flange		1	2.0	0.002	124.2	0.284	-1.30	-0.00465	0.000	0
(2x) Elbow, 1/8 not x 1/4 tube, push to connect		1	15.0	0.015	124.2	0.284	-2.30	-0.01465	0.000	0
Piston, Diaphragm	49811	1	135.3	0.1353	114.3	0.185	0.00	0.00835	0.000	0
Diaphragm, Internal (Bellofram)	52359	1	551.9	0.5519	115.7	0.199	0.00	0.00835	0.000	0
Sikaflex Adhesive Compound	55459	1	15.6	0.0156	114.3	0.185	0.00	0.00835	0.000	0
Cylinder Head, Diaphragm with extra port		1	336.4	0.3364	111.5	0.157	0.00	0.00835	0.000	0
Branch Tee, 1/4 tube x 1/8 NPT, push to connect; 1/4 tube plug		1	9.2	0.0092	109.8	0.140	-6.10	-0.05265	0.000	0
Elbow, 1/8 NPT M x 1/8 NPT M, brass	52426	1	26.9	0.0269	109.8	0.140	6.10	0.06935	0.000	0
Kepner Check Valve 404A-1-200		1	26.6	0.0266	109.8	0.140	2.00	0.02835	0.000	0
90 Elbow, 1/8 npt x 1/8 tube, push to connect		1	7.4	0.0074	109.8	0.140	-0.30	0.00535	0.000	0
Filter, Baston (25 micron) / with zip tie		1	20.9	0.0209	109.8	0.140	2.00	0.02835	0.000	0
Union, 1/4 tube, push to connect		1	1.8	0.0018	109.8	0.140	2.00	0.02835	0.000	0
90 Elbow, 1/8 npt x 1/4 tube, push to connect		1	7.5	0.0075	109.8	0.140	2.00	0.02835	0.000	0

Appendix E: 0% rudder										
SEAGLIDER - TRIM AND BALANCE		SG 194	Date	24-Oct-13						
	Vehicle CG	x_CG	95.752		cm					
		z_CG	-0.835		cm					
		y_CG	0		cm					
Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Bleed screw, 10-32 x .125 PHMS with seal	55477	1	1.8	0.0018	109.8	0.140	7.60	0.08435	0.000	0
Cylinder, Diaphragm	49812	1	421.5	0.4215	115.1	0.193	0.00	0.00835	0.000	0
(8x) 5-40 x 3/8 SHCS + #5 LW, head to cylinder	55251	1	7.5	0.0075	110.6	0.148	0.00	0.00835	0.000	0
Standoff (starboard)	55299	1	12.2	0.0122	121.3	0.255	6.60	0.07435	0.000	0
Standoff (port)	55299	1	12.5	0.0125	121.3	0.255	-6.60	-0.05765	0.000	0
(4x) 6-32 x 3/8 SHCS, standoff to cylinder	55183	1	4.5	0.0045	120.7	0.249	0.00	0.00835	0.000	0
(2x) Saddles, mounting, Aft PCB + (4x) 4-40 x 1/4 PHMS	52373 / 55257	1	5.3	0.0053	115.6	0.198	8.10	0.08935	0.000	0
(2x) Saddles, mounting, C PCB + (4x) 4-40 x 1/4 PHMS	49888 / 55257	1	6.7	0.0067	115.6	0.198	3.80	0.04635	0.000	0
(2x) Saddles, mounting, T PCB + (4x) 4-40 x 1/4 PHMS	49889 / 55257	1	6.7	0.0067	115.6	0.198	-3.80	-0.02965	0.000	0
(2x) Saddles, mounting, SBE O2 PCB + (4x) 4-40 x 1/4 SHCS	49890 / 55257	1	4.5	0.0045	115.6	0.198	-8.10	-0.07265	0.000	0
Linear Poteniometer A w/ cable assy (W21), Linear pot clamp, 3-48 x 1/2 SHCS, Spring, Linear pot rod end	52255 / 49814 / 55220	1	40.9	0.0409	121.3	0.255	6.40	0.07235	0.000	0
Linear Potentiometer B w/ cable assy (W20), Linear pot clamp, 3-48 x 1/2 SHCS, Spring, Linear pot rod end	52256 / 49814 / 55220	1	38.5	0.0385	121.3	0.255	6.40	0.07235	0.000	0
(2x) 5-40 x 1/2 SHCS + #5 LW, Clamp to cylinder	55250 / 55212 / 55225	1	1.9	0.0019	121.0	0.252	0.00	0.00835	0.000	0
(2x) 6-32 x 1/2 SHCS + #6 LW, boost pump bracket to endcap	55183 / 55217	1	2.7	0.0027	122.6	0.268	0.00	0.00835	0.000	0
(4x) 6-32 x 1/2 SHCS + #6 LW, cylinder standoff to endcap	55184 / 55217	1	5.4	0.0054	121.7	0.259	0.00	0.00835	0.000	0
Aft Terminal PCB	55302	1	25.0	0.025	115.6	0.198	8.60	0.09435	0.000	0
SBE Conductivity PCB (included with sensors)	55385	1	32.6	0.0326	115.6	0.198	4.30	0.05135	0.000	0
Conductivity, PCB jumper, CA W26	52261	1	3.1	0.0031	112.8	0.170	3.80	0.04635	0.000	0
Shield, Radiation, C PCB + (4x) 2-56 x 1/4 BHCS	49887 / 55221	1	20.1	0.0201	115.6	0.198	4.80	0.05635	0.000	0
SBE Temperature PCB (included with sensors)	55388	1	22.0	0.022	115.6	0.198	-4.30	-0.03465	0.000	0
Temperature, PCB jumper, CA W25	52260	1	5.2	0.0052	112.8	0.170	1.30	0.02135	0.000	0
Shield, Radiation, T PCB + (4x) 2-56 x 1/4 BHCS	52371 / 55221	1	16.5	0.0165	115.6	0.198	-4.80	-0.03965	0.000	0
Oxygen, PCB jumper, CA W33 (only used with SBE 43f)	52265	1	6.1	0.0061	112.8	0.170	0.00	0.00835	0.000	0
Ribbon Cable (W13) (gray)	52248	1	29.6	0.0296	98.1	0.023	8.90	0.09735	0.000	0
Ribbon Cable (W14) (rainbow)	52249	1	37.1	0.0371	98.1	0.023	8.90	0.09735	0.000	0
CT Sensor Sail Assembly (stanchion, guard, nut, nacelle, (6x) 6-32 x 5/16 FHMS, o-ring (E70-016), o-ring (E70-012), SBE components)	52368	1	293.8	0.2938	121.6	0.258	17.90	0.18735	0.000	0
Mass Shifter Stability Rod Assy w/ Hard Stop ("the stinger")		1	87.6	0.0876	111.5	0.157	0.00	0.00835	0.000	0
Electronic rail, port	49846	1	60.9	0.0609	59.7	-0.361	0.00	0.00835	0.000	0
Electronics rail, starboard	52379	1	60.9	0.0609	59.7	-0.361	0.00	0.00835	0.000	0
Seaglider Mainboard (Mother board, TT8, CF8, flash card)	522990-B.4	1	263.5	0.2635	54.6	-0.412	0.70	0.01535	0.000	0
Insulation Sheet-OEM/Main	52370	1	4.0	0.004	55.2	-0.406	0.44	0.01275	0.000	0
(6x) 6-32 x 1/4 PHMS, main board to rails	55237	1	4.5	0.0045	54.6	-0.412	0.50	0.01335	0.000	0

[illegible]

SEAGLIDER - TRIM AND BALANCE	SG 194	Date	24-Oct-13							
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Date

24-Oct-13							
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x_CG

cm

-0.835

cm

0

cm

Description	Part #	Qty	Total Weight grams	Total Weight kg	Long. CG (x) cm	Delta_x m	VCG&B (z) cm	Delta_z m	Lat. CG (y) cm	Delta_y m
Assy, OEM Navigation Board (transponder PCB, transformer/inductor, capacitors, GPS reciever, W6 GPS serial interface, GPS antenna cable, DC block, Cable Assembly W7, all soldered on wiring) + RAFOS clock and receiver board & cable to mainboard		1	559.1	0.5591	54.6	-0.412	-0.43	0.00405	0.000	0
(6x) 6-32 x 1/4 PHMS, OEM board to rails	55237	1	4.5	0.0045	54.6	-0.412	-0.43	0.00405	0.000	0
Assy, Iridium Moedm and bracket (includes satellite reciever - iridium LBT, cable assy W3, all clamps for wires, RF switch, DC block, and mounting hardware)	52324	1	271.6	0.2716	54.9	-0.409	1.90	0.02735	0.000	0
4x) 6-32 x 1/4 PHMS + #6 LW, modem cage to rails	55209 / 55217	1	3.4	0.0034	52.8	-0.430	0.29	0.01125	0.000	0
Battery Pack, LV w/ CA W1	52237	1	2664.4	2.6644	54.9	-0.409	-1.81	-0.00975	0.000	0
Battery Pack Cage, LV	49847	1	71.6	0.0716	54.9	-0.409	-1.64	-0.00805	0.000	0
(6x) 6-32 x 1/4 PHMS + #6 LW, batt. cage to rails	55209 / 55217	1	5.4	0.0054	55.3	-0.405	-0.25	0.00585	0.000	0
Hull, Fwd Battery	49817	1	3646.0	3.646	110.6	0.148	0.00	0.00835	0.000	0
Hull, Aft Battery	49817	1	3638.6	3.6386	85.5	-0.103	0.00	0.00835	0.000	0
Joint Ring, battery hulls	49818	1	674.7	0.6747	98.9	0.031	0.00	0.00835	0.000	0
(8x) 6-32 x 3/4 SHCS + #6 LW, joint ring to fwd/aft batt. Hull	55214 / 55217	1	13.3	0.0133	98.1	0.023	0.00	0.00835	0.000	0
(2x) O-Ring, E70-271	52412	1	17.1	0.0171	98.1	0.023	0.00	0.00835	0.000	0
(12x) Clips, Rib	49883	1	25.4	0.0254	98.1	0.023	3.95	0.04785	0.000	0
(12x) Clips, Wiring	52405	1	18.7	0.0187	98.1	0.023	3.95	0.04785	0.000	0
Oil Absorption pad, 1.9" x 10.125" + (2x) rib clips	55527 / 49883	1	11.2	0.0112	98.1	0.023	3.95	0.04785	0.000	0
Endcap, Forward	49830	1	1897.1	1.8971	39.5	-0.563	0.00	0.00835	0.000	0
Acoustic transducer (ITC-3013), w/ cable W36	52267 / 52223	1	1438.7	1.4387	22.9	-0.729	0.00	0.00835	0.000	0
O-Ring, E70-238, transducer to Endcap	52413	1	3.2	0.0032	26.7	-0.691	0.00	0.00835	0.000	0
(6x) Alum 1/4-20 x 3/4 SHCS + 1/4 LW, xducer to end cap	55236 / 55252	1	15.2	0.0152	27.2	-0.686	0.00	0.00835	0.000	0
Hull, Electronics	49829	1	1696.0	1.696	58.0	-0.378	0.00	0.00835	0.000	0
O-Ring, E70-263, elec hull to fwd endcap	52410	1	7.1	0.0071	50.2	-0.456	0.00	0.00835	0.000	0
(8x) 6-32 x 1/2 SHCS + #6 LW, electric hull to fwd encap	55183 / 55217	1	10.3	0.0103	50.5	-0.453	0.00	0.00835	0.000	0
Compass bracket	49848	1	24.4	0.0244	30.5	-0.653	-0.35	0.00485	0.000	0
Compass w/compass carrier PCB	55305 / 52466	1	46.8	0.0468	30.5	-0.653	0.00	0.00835	0.000	0
(4x) 4-40 x 3/8 SHCS + #4 x 1/8L stand off	55256 / 55282	1	2.6	0.0026	30.5	-0.653	-0.31	0.00525	0.000	0
(2x) 4-40 x 3/8 SHCS + #4 LW, comp mt to fwd encap	55228 / 55233	1	1.3	0.0013	27.4	-0.684	-0.81	0.00025	0.000	0
Cable Assembly W9 (compass cable)	52244	1	21.5	0.0215	38.9	-0.569	0.38	0.01215	0.000	0
Total weight (SUM of components' weight col D)			44641	44.704	Excluding movable battery mass					
Total Scale Weight			58852							